Rayfield-Based Calibration and Effective Physical Model Identification of a Common Main Objective Stereo Microscope

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Abstract

Common Main Objective (CMO) stereo microscopes violate the single-optical-centre assumption of standard camera calibration, making the joint estimation of optical parameters and target poses strongly ill-conditioned. We propose a two-stage calibration strategy: a Zernike rayfield—one 3D line per pixel—is first fitted to ChArUco corner observations, then used as a diagnostic observable to identify a compact physical model by residual modal analysis. On a real CMO instrument from the open Pycaso dataset, the resulting 26-parameter model—a quasi-telecentric CMO skeleton with per-channel SE(3) arm alignment—reaches $1.06\,\mathrm{px}$ reprojection error, whereas standard OpenCV stereo calibration exceeds $300\,\mathrm{px}$. Used to initialise a free-pose bundle adjustment stabilised by a Schur-spectrum observability prior, the same model refines to $0.28\,\mathrm{px}$—within $0.1\,\mathrm{px}$ of an interpretation-free Soloff polynomial floor ($0.18\,\mathrm{px}$)—while retaining a physically interpretable parameterisation whose geometric descriptors are effective (gauge-dependent) rather than absolute. Profilometry of an independent specimen shows that free-pose Zernike reconstruction over-amplifies the relief because of a missing pose-metrology prior rather than an intrinsic rayfield limitation; supplying the nominal translation-stage ladder as a stage-metric observability prior restores agreement with the profilometric reference to within $\approx 10\,\%$, although traceable accuracy on a certified 3D artefact remains future work. The rayfield therefore acts not only as a calibration model but as a diagnostic observable that guides compact physical-model construction and stabilises its subsequent refinement.

Keywords: stereo microscope; CMO; non-central calibration; Zernike rayfield; BIC model selection; telecentricity

1 Introduction

Stereo calibration is a mature technology when the camera obeys the pinhole model: all chief rays pass through a single optical centre, and mature toolchains (OpenCV, MATLAB, Bouguet) produce reliable results. This assumption breaks down for Common Main Objective (CMO) stereo microscopes, which use a single large objective shared between two off-axis sub-apertures [11, 23]. Each channel's chief rays appear to originate from a different effective sub-pupil, separated by the stereo baseline (Figure 1). OpenCV's `stereoCalibrate` [18] assumes one optical centre per camera; on the CMO architecture it produces unusable results ($>$ 300 px reprojection error under the tested configuration).

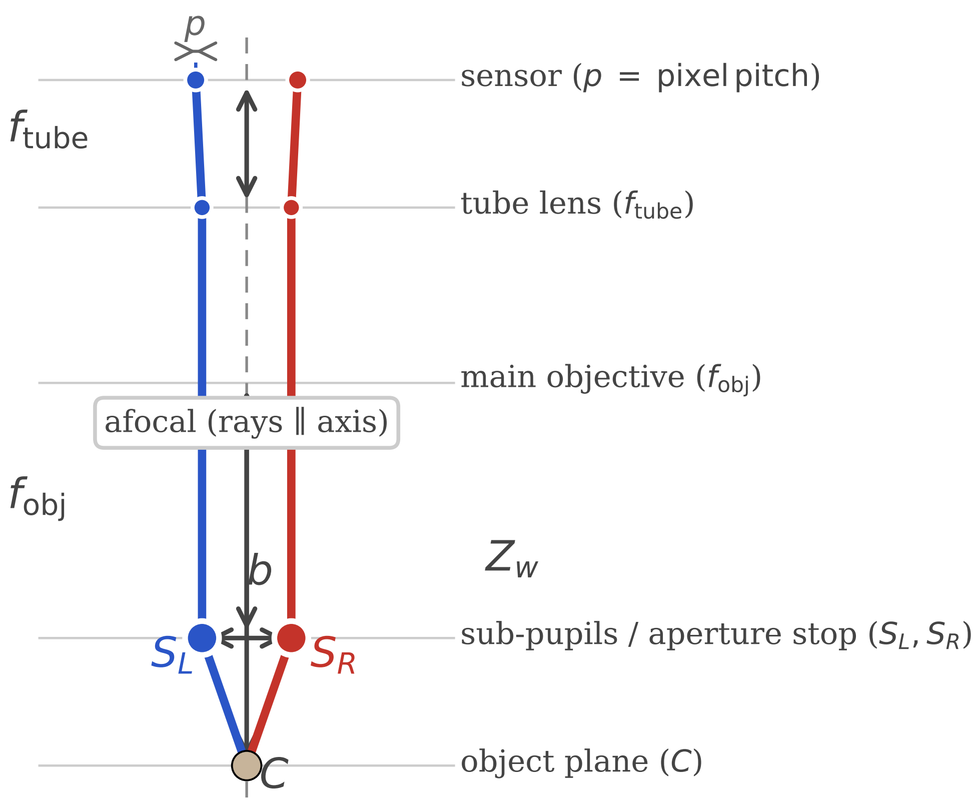

Figure 1. CMO stereo microscope optical layout. Two off-axis sub-pupils share a single main objective. Chief rays converge toward the object plane through different effective origins—not through a single common centre.

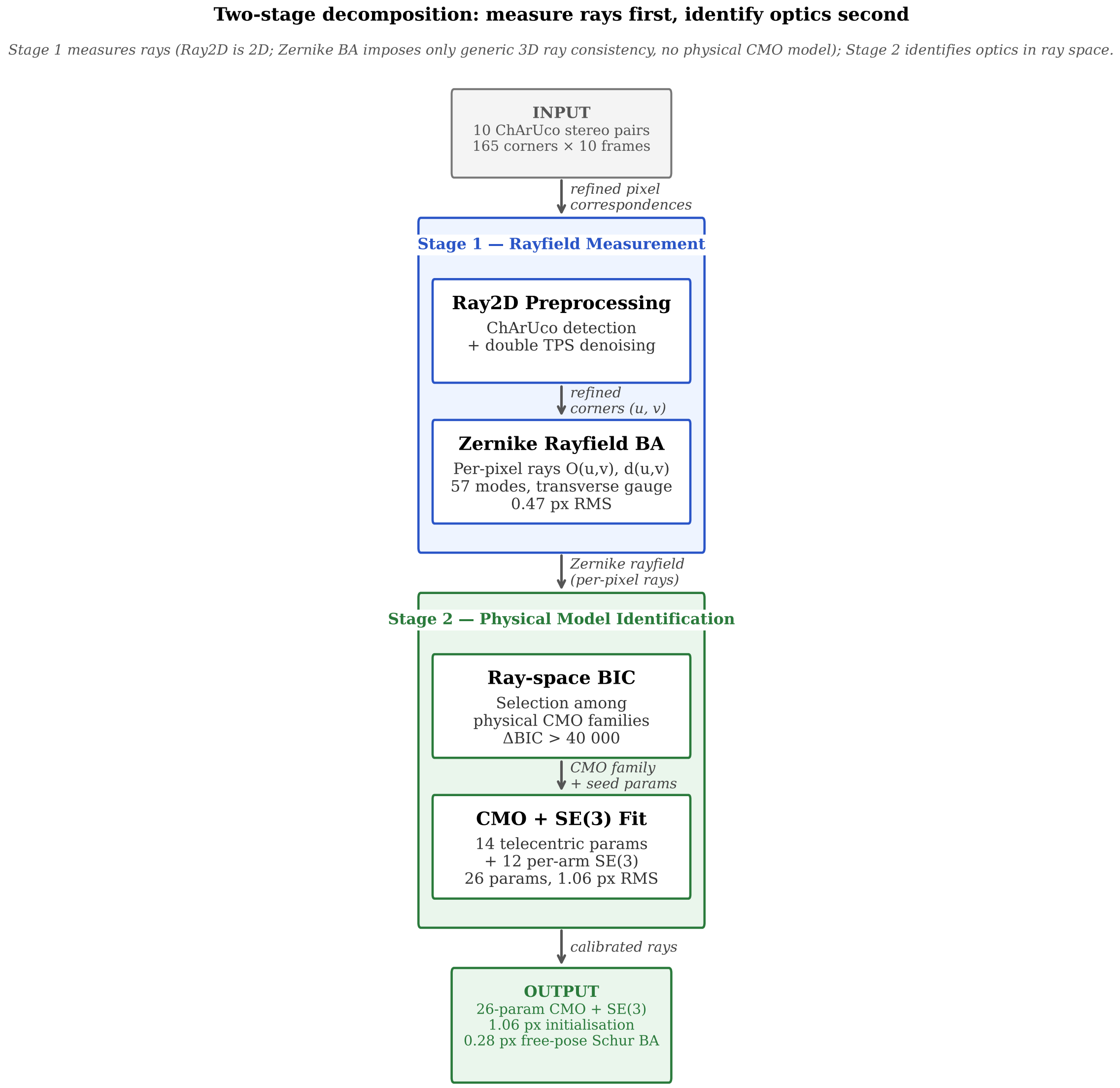

Figure 2. Pipeline overview. The two-stage decomposition separates rayfield measurement (Ray2D preprocessing + Zernike BA) from physical model identification (ray-space BIC selection). The Ray2D stage is purely 2D and does not impose a 3D camera model.

The broader problem is that many optical instruments—CMO and Greenough stereo microscopes, endoscopes, Scheimpflug cameras, multi-camera arrays—depart from the central projection model. Among these, the CMO architecture is particularly challenging: in a Greenough microscope the two independent objectives yield two well-separated optical centres that can be modelled as individual perspective cameras, whereas a CMO microscope shares a *single* large objective between off-axis sub-apertures, producing decentred exit pupils whose chief rays do not converge to a common point—a subtle geometric coupling that violates the pinhole assumption far more severely than a simple dual-perspective system. Generic camera models [10, 30, 25] and recent stereo microscope calibration studies [26, 35, 14] highlight the need for non-central approaches, but do not provide a systematic method for identifying the correct physical model family from observations (see Section 2 for a detailed review).

StereoComplex [33] addresses this gap: it fits a Zernike rayfield $\mathcal{R}(u, v) = (O(u, v), d(u, v))$—a 3D line for every pixel—to corner observations, using a Zernike polynomial basis [34, 17]. This two-stage decomposition—measure rays first, identify optics second (Figure 2)—is the central insight: it decouples the severely ill-conditioned joint estimation of optical parameters and board poses into a well-conditioned rayfield fit followed by pose-free model selection [25]. From this *measured* rayfield, physical descriptors can be read directly (Section 3), structural mismatches diagnosed by residual modal analysis (Section 3.3), and compact physical models constructed iteratively (Section 3.4–3.5).

This paper presents the first real-data validation of the StereoComplex rayfield-mediated non-central calibration workflow on a CMO stereo microscope using legacy ChArUco [9, 22] images from the open Pycaso dataset [8]. To demonstrate the robustness of the proposed diagnostic and calibration pipeline, we intentionally benchmark it against the open-source Pycaso dataset. Since this dataset consists exclusively of standard planar target poses—which are known to introduce strong depth-scale ambiguities in non-central systems—it serves as a stringent test case: we show that the analytical framework successfully decouples the structural optics from the pose estimation and recovers the missing physical degrees of freedom without requiring custom-made 3D metrology gauges or modified acquisition protocols. The contributions are:

1. A complete pipeline from raw ChArUco images to a compact 26-parameter physical CMO model, validated on real data. The model provides *effective* identification: the shear coefficients and telecentric skeleton are data-supported, while geometric scale parameters (WD, b, f_{obj}) and most SE(3) components are absorbed by the rayfield gauge and pose coupling (see §3.7 for a per-parameter identifiability analysis).

2. A residual analysis method—Zernike modal decomposition of direction and moment differences between a candidate physical model and the measured rayfield—that identifies missing degrees of freedom.

3. A two-level model-selection framework: ray-space BIC is used to identify the optical family, while a separate operational reprojection guard rejects models whose pixel RMS is not usable for calibration.

4. Demonstration that the Zernike rayfield serves a *double role* in direct bundle adjustment: (i) as an initialiser, it places the BA in the correct convergence basin, improving pixel RMS from $1.06\,\mathrm{px}$

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to $0.88\,\mathrm{px}$ ($17\,\%$) while leaving all 26 model parameters unchanged to within 10^{-3}; (ii) as an *observability prior*, the Schur complement of the Fisher information matrix at the rayfield solution defines an anisotropic penalty that suppresses optics–pose compensation—the prior reduces weak-mode parameter drift by a factor of $280\times$ (from 0.93 to 0.0033) at a cost of only $0.038\,\mathrm{px}$ in 2D reprojection RMS. In the free-pose Schur-stabilised BA, the 26-parameter physical model reaches $0.28\,\mathrm{px}$—within $0.1\,\mathrm{px}$ of the Soloff polynomial floor—with its optics held in the observable subspace rather than drifting into the unobservable directions that the unregularised optimum ($0.24\,\mathrm{px}$) exploits.

5. A stage-metric observability prior for Zernike rayfield bundle adjustment, showing that the apparent dense-depth over-amplification observed with near-parallel planar Pycaso data is caused by missing axial pose-metrology information rather than by an intrinsic limitation of non-central Zernike rayfields.

2 Related Work

2.1 Generic and Non-Central Camera Calibration

The pinhole camera model with polynomial distortion [36, 5] remains the standard in computer vision toolkits [18, 4]. For wide-angle and fisheye lenses, unified models [15, 12, 24] extend the pinhole paradigm with additional radial terms. However, these parametric models share a fundamental assumption: all chief rays pass through a single optical centre.

For cameras that violate this assumption, generic (non-parametric) models assign a per-pixel ray without imposing a central projection constraint. Grossberg and Nayar [10] introduced the raxel model, treating each pixel as an independent ray. Sturm and Ramalingam [30], later refined by Ramalingam and Sturm [20], developed a generic calibration framework from multiple views of a planar target, showing that a flexible pixel-to-ray mapping can be estimated without knowing the camera model *a priori*. Miraldo and Araujo [16] proposed smooth generic camera models using continuous ray-field interpolation. Soloff et al. [28] introduced polynomial expansions mapping 3D coordinates to 2D pixel positions; this approach, widely adopted in stereo-microscope calibration [8], treats the camera as a black-box regression problem and can achieve excellent pixel accuracy—as we confirm in §4.6, a degree-3 Soloff fit reaches $0.18\,\mathrm{px}$ RMS on the Pycaso dataset with 80 parameters. Camposeco et al. [7] demonstrated spline-based generic calibration with subpixel accuracy. Ramalingam, Sturm, and Nayar [21] unified these approaches in a generic structure-from-motion framework.

Most recently, Schöps et al. [25] showed that highly parameterised generic models (10,000+ parameters) can outperform classical 12-parameter models on standard benchmarks, challenging the assumption that parametric models are inherently more robust.

Our work differs from these generic approaches in two ways: (1) we use the Zernike rayfield not as a final calibration model but as a *diagnostic instrument* to discover the correct physical model family, and (2) we construct a compact, interpretable physical model guided by residual structure rather than imposing a generic non-parametric fit. Importantly, we do not claim superior pixel accuracy—the Soloff polynomial is more accurate at equivalent parameter counts (§4.6)—but superior physical interpretability: unlike a black-box polynomial, the 26-parameter model has a physically interpretable parameterisation (baseline, working distance, convergence angle, shear, SE(3) arm poses), although only some combinations are independently identifiable after pose marginalisation (§3.7). Interpretability is, moreover, not the rayfield's only advantage over a polynomial regressor: although the Soloff fit achieves lower *in-sample* pixel RMS, the rayfield's 3D ray constraint acts as a *structural regulariser*. On held-out extreme poses the degree-3 Soloff fit degrades by $4.65\times$ while the 180-parameter Zernike rayfield degrades by only $1.86\times$ (§4.6), because each pixel's ray must stay geometrically consistent with a rigid 3D board rather than memorising individual training poses. The residual accuracy gap is, moreover, small in absolute terms: after the free-pose Schur-stabilised refinement the 26-parameter model reaches $0.28\,\mathrm{px}$, within $0.1\,\mathrm{px}$ of the Soloff floor (§4.6).

2.2 Stereo Microscope Calibration

Stereo microscopes pose a specific calibration challenge: the two optical channels share a common main objective but have separate effective pupils [26]. Digital image correlation under optical microscopes requires careful distortion correction [35, 14].

These methods calibrate a specific instrument but do not address the generic problem of *identifying* the correct optical model family from observations. Our contribution is a systematic framework that measures the ray-field first, then uses residual analysis to construct a physical model that matches the observed structure—here, discovering the telecentric CMO family and the SE(3) arm misalignment.

2.3 Model Selection in Camera Calibration

Model selection in calibration traditionally relies on reprojection RMS, which penalises underfitting but is insensitive to overfitting. The Akaike Information Criterion (AIC) [1] and Bayesian Information Criterion (BIC) [27] add parameter-count penalties, providing a principled trade-off between fit quality and model complexity. Strecha et al. [29] established benchmark datasets and evaluation protocols for camera calibration and multi-view stereo. Our work extends BIC-based model selection to ray space: we compare optical model families (pinhole, Brown-Conrady, parallel-plate, telecentric CMO) against a measured Zernike rayfield, and add an operational reprojection guard that rejects models whose pixel RMS exceeds a usability threshold—a constraint that the raw ray-space BIC does not capture.

2.4 Plücker Coordinates and Line-Based Geometry

Plücker coordinates [19] represent a 3D line as a 6-vector (d, m) where d is the direction and $m = O \times d$ is the moment. They are widely used in geometric computer vision for line-based structure from motion and calibration [2, 31]. Our residual analysis uses Plücker moments to separate direction errors from origin errors: Δd reveals angular misalignment; Δm reveals the effective ray origin shift. The Z_0^0-dominated residual in both Δd and Δm is the key diagnostic that guides the SE(3) arm alignment.

2.5 Residual-Guided Model Construction

The idea of using residuals to guide model refinement has roots in adaptive optics and wavefront sensing [34, 17], where Zernike decomposition of a distorted wavefront identifies specific aberrations (defocus, astigmatism, coma) that are then corrected by deformable mirrors. Our method applies the same principle to camera calibration: the Zernike decomposition of the residual between a candidate physical model and the measured rayfield identifies which physical degree of freedom is missing—a Z_0^0-dominated residual points to a global offset (SE(3) arm misalignment), as we verify here. The analogy guides interpretation but is not a strict optical identification: a second-order-dominated residual would be *consistent* with low-order field-dependent aberrations such as field curvature or astigmatic components, but pinning down a unique optical cause would require a complementary pupil-space or wavefront-level analysis that we do not perform. This is distinct from generic non-parametric approaches [25] that fit a flexible model without extracting physical meaning from the residual structure.

3 Method

3.1 Zernike Order Selection by BIC

The Zernike rayfield has two hyperparameters: the origin polynomial order o and the direction polynomial order d. Increasing either adds Zernike modes, reducing pixel RMS at the cost of more parameters. We sweep $o \in \{0, 1, 2\}$ and $d \in \{2, 3, 4\}$ (seven selected combinations) and select the optimal order by BIC [27] (Figure 3):

$$\mathrm{BIC}_{\mathrm{px}} = k \log(N_{\mathrm{obs}}) + N_{\mathrm{obs}} \log(\mathrm{RSS}_{\mathrm{px}}/N_{\mathrm{obs}}), \tag{1}$$

where k is the total number of parameters (Zernike coefficients + poses), $N_{\mathrm{obs}} = 3300$ corner observations, and $\mathrm{RSS}_{\mathrm{px}}$ is the sum of squared pixel-equivalent residuals. The BIC treats these 3300 corners as independent Gaussian observations; corners within a frame are in fact spatially correlated, so the absolute BIC magnitudes are optimistic. The *family*-level selection used here is nonetheless robust: the gaps between optical families (ΔBIC $> 40\,000$) dwarf any plausible reduction in the effective sample size.

Unless stated otherwise, parameter counts reported in BIC tables (e.g. Table 1) include both rayfield coefficients and calibration-pose parameters. Counts used when discussing extrapolated rayfield capacity, such as the “180-parameter O(4)+d(4) Zernike rayfield” in §4.6, refer to rayfield coefficients only and exclude the pose variables.

Table 1. Zernike order sweep: pixel RMS and BIC.

Model	k	RMS (px)	$\mathrm{BIC}_{\mathrm{px}}$
O(0)+d(2)	57	0.470	−4522
O(1)+d(2)	69	0.444	−4803
O(0)+d(3)	81	0.419	−5080
O(1)+d(3)	**93**	**0.412**	**−5093**
O(2)+d(3)	111	0.409	−4995
O(1)+d(4)	123	0.412	−4858
O(2)+d(4)	141	0.410	−4743

Table 1 summarises the full sweep. The order sweep is not a full 3×3 grid over all origin and direction orders; it reports the seven combinations retained for physical CMO model identification. The BIC minimum is therefore interpreted as a diagnostic of the best flexible rayfield accuracy available in this data regime. The O(0)+d(2) rayfield is not claimed to be the statistical BIC optimum; it is deliberately retained as the physically consistent reference for CMO model construction because it is the lowest-order model consistent with a rigid-sub-pupil CMO skeleton, and it remains within 0.06 px of the BIC-minimum RMS. The later specimen and stage-prior experiments (§4.8, §4.8.1) deliberately use a field-resolved O(2)+d(2) Zernike rayfield (72 rayfield coefficients plus 15 constrained-pose parameters, 87 total) to test whether dense-depth errors originate from axial pose metrology rather than from the rigid-origin restriction of the diagnostic O(0)+d(2) rayfield.

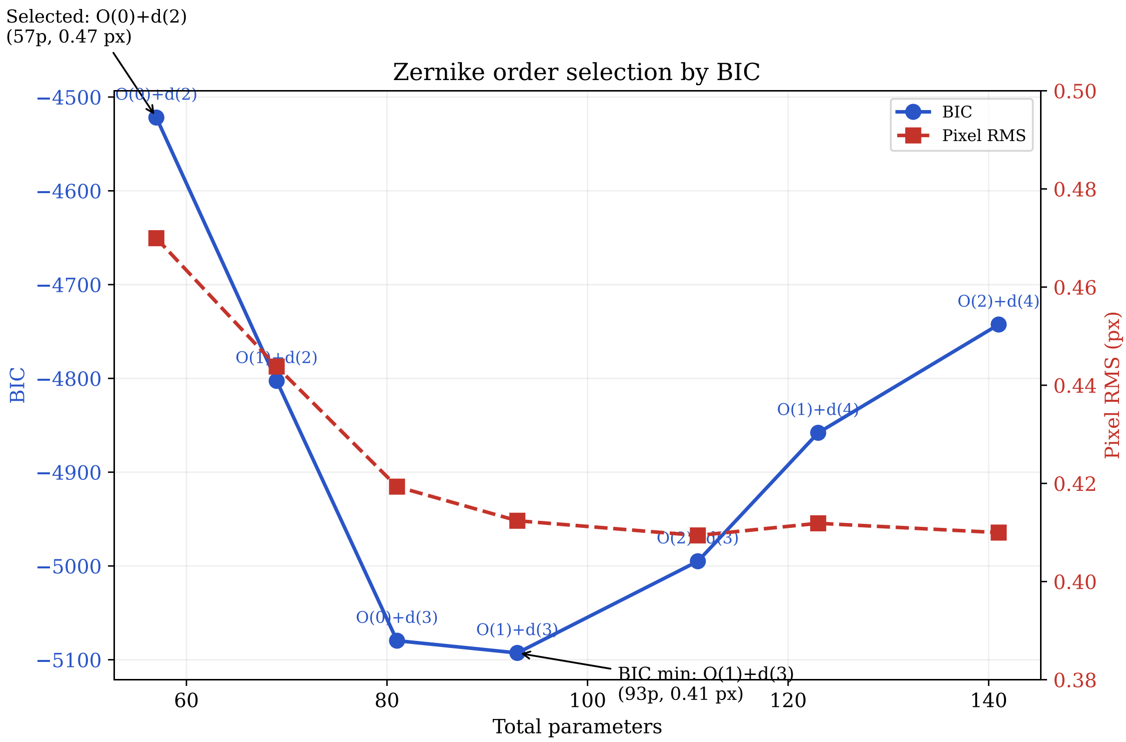

Figure 3. Zernike order selection by BIC. Pixel RMS falls from 0.47 px at 57 params to 0.41 px near the BIC minimum (93 params) and then saturates within corner-detection noise (it is not strictly monotonic beyond, varying by < 0.005 px up to 141 params), but BIC penalises complexity. The BIC minimum is at O(1)+d(3) (93 params, BIC = -5093, RMS = 0.41 px), though O(0)+d(2) (57 params, BIC = -4522) is within $\sim$10 % in RMS at 40 % fewer parameters. The O(0)+d(2) model (57 params, 0.47 px) is selected for physical model construction because it is internally consistent with the rigid-sub-pupil CMO assumption, achieves $90\,\%$ of the noise-floor accuracy at nearly half the parameters of the BIC minimum, and the marginal improvement above d(2) falls within corner detection noise.

The BIC minimum at O(1)+d(3) achieves $0.41\,\mathrm{px}$ RMS with 93 parameters. We nevertheless select O(0)+d(2) (57 params, $0.47\,\mathrm{px}$) as the preferred rayfield configuration, for three reasons:

1. **Simplicity:** 57 params vs. 93 — nearly half the parameters for a penalty of only $0.06\,\mathrm{px}$ in RMS ($\sim$0.1 px in practice).

2. **Marginal improvement:** the residual gain from d(2) to d(3) ($0.47\,\mathrm{px} \rightarrow 0.41\,\mathrm{px}$) is well within typical subpixel noise of ChArUco detection, so O(0)+d(2) is retained as the physically consistent reference even though O(1)+d(3) is the statistical BIC minimum.

3. **Consistency with the physical model:** the CMO physical model assumes rigid sub-pupils ($o = 0$). Using the same origin order in the Zernike rayfield makes the two representations internally consistent and comparable: the Zernike field measures what the physical model can represent.

Selecting O(0)+d(2) achieves $90\,\%$ of the noise-floor accuracy with a representation that is internally consistent with the physical model it validates.

It is worth stating what the discarded origin order would mean physically. An $o = 1$ origin field lets the effective sub-pupil position drift linearly across the field of view—a field-dependent *pupil walk*, the optical signature of a residual pupil aberration or secondary telecentricity error. The marginal RMS gain it brings, however, stays within ChArUco corner-detection noise (Table 1), so on this data a genuine pupil walk cannot be distinguished from fit noise; we therefore retain the rigid-sub-pupil $o = 0$ model and read the $o = 1$ preference of the BIC sweep as, at most, a hint of higher-order pupil behaviour that lower-noise corners or an independent pupil measurement would be needed to confirm.

3.2 Per-Pixel Ray-Field and Transverse Gauge

For each pixel (u, v) on the sensor, the imaging system defines a 3D line in space. We represent this line $\mathcal{L}_c(u, v)$ by an origin point $O_c(u, v) \in \mathbb{R}^3$ and a unit direction vector $d_c(u, v) \in \mathbb{S}^2$, where $c \in \{L, R\}$ denotes the left or right channel. Points on the line are parameterised as $O_c + t\, d_c$ for $t \in \mathbb{R}$.

The origin $O_c(u, v)$ is not unique: any point along the ray would represent the same line. We fix this gauge freedom by enforcing the **transverse gauge** $O_c \cdot d_c = 0$, which selects the point on the ray closest to the coordinate origin. This eliminates one redundant degree of freedom per ray and stabilises the parameterisation.

The origin and direction fields are expanded on a Zernike polynomial basis [34, 17]:

$$O_c(u, v) = \sum_m a^O_{c,m}\, Z_m(\tilde{u}, \tilde{v}), \qquad d_c(u, v) = \mathrm{normalise}\left(d_c^{\mathrm{pinhole}}(u, v) + \sum_{m \le d_{\max}} a^d_{c,m}\, Z_m(\tilde{u}, \tilde{v}) \right), \tag{2}$$

where Z_m are the real-valued Zernike polynomials on the unit disk, $\tilde{u}, \tilde{v}$ are the corresponding in-disk normalised pixel coordinates (the normalisation is specified below), d_c^{pinhole} is the initial pinhole direction from a fixed central projection matrix K, and the direction perturbation is projected transverse to d_c^{pinhole} before renormalisation. The expansion order for the origin field is $o_{\max}$ and for the direction field is $d_{\max}$; selecting $o_{\max} = 0$ and $d_{\max} = 2$ (see §3.1) gives 3 origin coefficients and 18 direction coefficients per channel. In geometrical (eikonal) optics, a ray direction in a homogeneous isotropic medium such as air can be interpreted locally as the normal to an eikonal wavefront. We therefore read the fitted unit vector $d_c(u, v)$ as an *effective chief-ray direction*—equivalently, a local wavefront-normal direction in back-projected object space—and *not* as a direct measurement of optical phase. Consistently, the Zernike expansion of its perturbation $\sum_m a^d_{c,m} Z_m$ should be read as a smooth modal description of transverse ray-direction deviations, analogous to the modal language of classical wavefront aberration theory, rather than as an interferometric wavefront reconstruction; note in particular that these modes are defined here over the image/sensor domain, not over the pupil for a fixed field point, and need not derive from an integrable scalar wavefront. Each Z_m in the sums above is a real-valued Zernike polynomial $Z_n^m(\rho,\theta) = R_n^m(\rho)\,\Theta(\theta)$, with angular factor $\Theta \in \{1, \cos m\theta, \sin m\theta\}$, taken in the *unnormalised* amplitude convention $R_n^m(1) = 1$—no RMS or OSA/ANSI rescaling, so each fitted coefficient is the mode's peak amplitude (mm for the origin field, dimensionless for the direction perturbation)—and ordered canonically by increasing radial order n, then azimuthal order m, cosine before sine. Appendix B states the radial polynomial R_n^m, the mode table for the orders used here, and the exact source reference [33]. The radial coordinate is normalised by the sensor *diagonal* (a circumscribed-circle scheme): pixel (u, v) maps to $r = \|(2u/(W-1)-1,\ 2v/(H-1)-1)\|/\sqrt{2}$, so the four image corners fall exactly on $r = 1$ and every pixel satisfies $r \le 1$. All evaluations therefore stay inside the unit disk, where the polynomials are bounded, avoiding the corner growth a box ($[-1, 1]^2$) normalisation would incur. Choosing a rotationally symmetric Zernike basis reflects the rotational symmetry of the optical train, the rectangular sensor being treated as a truncated observation window on that field.

3.3 Ray-Space Bundle Adjustment

The rayfield coefficients $a^O_{c,m}, a^d_{c,m}$ and the board poses $\{\eta_p\}_{p=1}^P$ are estimated jointly from N ChArUco corner observations. Each observation k links a pixel (u_k, v_k) on channel c_k to a known 3D board point X_k expressed in the board frame. The board pose $\eta_p = (R_p, t_p) \in \mathrm{SE}(3)$ transforms X_k to the camera frame: $X_k^{\text{cam}} = R_p X_k + t_p$.

The **ray-space residual** measures the perpendicular distance from the transformed board point to the ray $\mathcal{L}_{c_k}(u_k, v_k)$:

$$r_k = \|(X_k^{\text{cam}} - O_{c_k}) - ((X_k^{\text{cam}} - O_{c_k}) \cdot d_{c_k})\, d_{c_k}\|. \tag{3}$$

This is the norm of the component of $X_k^{\text{cam}} - O_{c_k}$ orthogonal to the ray direction d_{c_k}. The objective is the sum of squared residuals:

$$\min_{\{a_{c,m}\},\{\eta_p\}} \sum_k r_k^2, \tag{4}$$

solved by nonlinear least-squares. This joint optimisation of the optical (rayfield) parameters and the target poses is a global non-linear least-squares problem—the *bundle adjustment* of photogrammetry and computer vision [31], here expressed in ray space (point-to-ray distances) rather than in reprojected pixels.

For the CMO microscope, the baseline fit imposes the physically-motivated constraint that the board is mounted on a translation stage: all P frames share a common rotation R and XY translation (t_x, t_y), with one free axial coordinate $t_{z,i}$ per frame. This reduces the pose parameters from $6P$ to $3 + 2 + P = 15$ (for $P = 10$ frames: $60 \to 15$), making the 57-parameter O(0)+d(2) Zernike BA sufficiently conditioned for physical-model construction. However, this baseline uses the stage metadata only to initialise the poses; it does not transmit the measured inter-plane spacing to the final rayfield.

The specimen depth-gain diagnostic of §4.8 motivates a second, explicitly metric pose model. Let z_i^{nom} be the nominal stage coordinate read from the acquisition metadata and $\bar{z}^{\text{nom}}$ its mean. The frame translations are reparameterised as

$$t_i = t_0 + \mathbf{a}\left[s\left(z_i^{\text{nom}} - \bar{z}^{\text{nom}}\right) + \varepsilon_i\right], \qquad \|\mathbf{a}\| = 1, \tag{5}$$

where $t_0 \in \mathbb{R}^3$ is the common translation, $\mathbf{a}$ is the stage axis (two degrees of freedom), s is a common dimensionless scale, and ε_i is the frame-specific positioning error. The geometric likelihood is augmented by

$$\mathcal{L}_{\text{stage}} = \left(\frac{s-1}{\sigma_s}\right)^2 + \sum_{i=1}^{P}\left(\frac{\varepsilon_i}{\sigma_\varepsilon}\right)^2, \tag{6}$$

after normalising the ray residuals by their empirical component scale σ_r. This separates systematic stage-scale uncertainty (σ_s) from repeatability/jitter (σ_ε). The original free-$t_{z,i}$ model is recovered when the ε_i are effectively unconstrained and $\mathbf{a} = \hat{z}$; the fixed nominal ladder is the limit $\sigma_s, \sigma_\varepsilon \to 0$. Equation (5) is used only in the post-hoc observability experiment of §4.8.1; the rayfield used to construct the CMO family remains the baseline O(0)+d(2) fit above.

The ray-space BA differs from classical reprojection minimisation [36] in a fundamental way: it optimises the 3D geometric consistency of rays rather than the 2D reprojection error. This is the measurement stage of the two-stage decomposition (see Section 4.2).

3.4 The Zernike Rayfield as Observable

The fitted rayfield $\mathcal{R}_c(u, v) = (O_c(u, v), d_c(u, v))$ is used as an experimental observable. Because the Zernike rayfield is a non-central model, the BA that fits it carries a scale degeneracy: the pixel projection is invariant under simultaneous scaling of all lengths and the effective focal length f_x (see §4.7 for the linear f_x–WD relationship). Three independent constraints intersect to anchor the absolute scale. (1) The ChArUco board has known square size (0.3 mm); from the observed corner spacing Δu_{square} we obtain the ratio $f_x/WD = \Delta u_{\text{square}}/0.3\,\text{mm} \approx 395\,\text{px/mm}$. (2) The motorised Z-stage that acquired the Pycaso dataset displaces the board by a known physical step $\Delta Z_{\text{stage}} \approx 0.07\,\text{mm}$ per frame ($\sim$10 % accuracy); the measured disparity shift Δd between successive frames gives $f_x = \Delta d \cdot WD/\Delta Z_{\text{stage}}$. (3) the working distance documented for the Pycaso acquisition configuration is $\approx$65 mm [8]. (The bare Planapo 1.0× objective is specified at 61.5 mm [13]; the mounted/effective working distance of the assembled instrument differs, and that 61.5 mm datasheet value anchors a separate chief-ray consistency check in §4.5.) Solving the three constraints jointly yields $f_x = 25\,600\,\text{px}$ and $WD = 64.7\,\text{mm}$, consistent at the $\sim$1 % level with the documented configuration working distance. All geometric descriptors (b, WD, f_{obj}) inherit this scale and are therefore reported as *effective* quantities in the chosen gauge; an independent external reference would be required to convert them to traceable measurements (Section 5.2). The centre-pixel values, from which the effective sub-pupils O_L, O_R are reconstructed, are: $b = \|O_R - O_L\| = 24.9\,\text{mm}$ (effective stereo baseline in the rayfield gauge), $z_p = (|O_{L,z}| + |O_{R,z}|)/2 = 2.5\,\text{mm}$ (sub-pupil depth), $WD = 64.7\,\text{mm}$ (effective working distance, from mean pose Z; convention: object-plane Z minus optical-axis origin), $f_{\text{obj}} = WD - z_p = 62.2\,\text{mm}$ (effective axial length), $\Theta = \arccos(d_L \cdot d_R) = 22.6°$ (effective full convergence angle at the centre pixel).

3.5 Physical Model Construction by Residual Analysis

The core methodology is iterative: propose a candidate physical model, compute its residual against the Zernike rayfield, decompose the residual on Zernike modes, and use the dominant mode to identify the missing degree of freedom.

3.5.1 Perspective CMO (Baseline)

The simplest CMO model places a perspective camera behind each decentred sub-pupil $S_c = (\pm b/2, 0, WD - f_{\text{obj}})$. This predicts $d_y(u,v) \propto (v - c_y)$—a strong linear gradient. The measured d_y field is nearly constant across the sensor (range 0.079, mean +0.059), while the perspective model predicts a range of 0.232—a 3× discrepancy (Figure 4). The real system is **object-space telecentric**, not perspective.

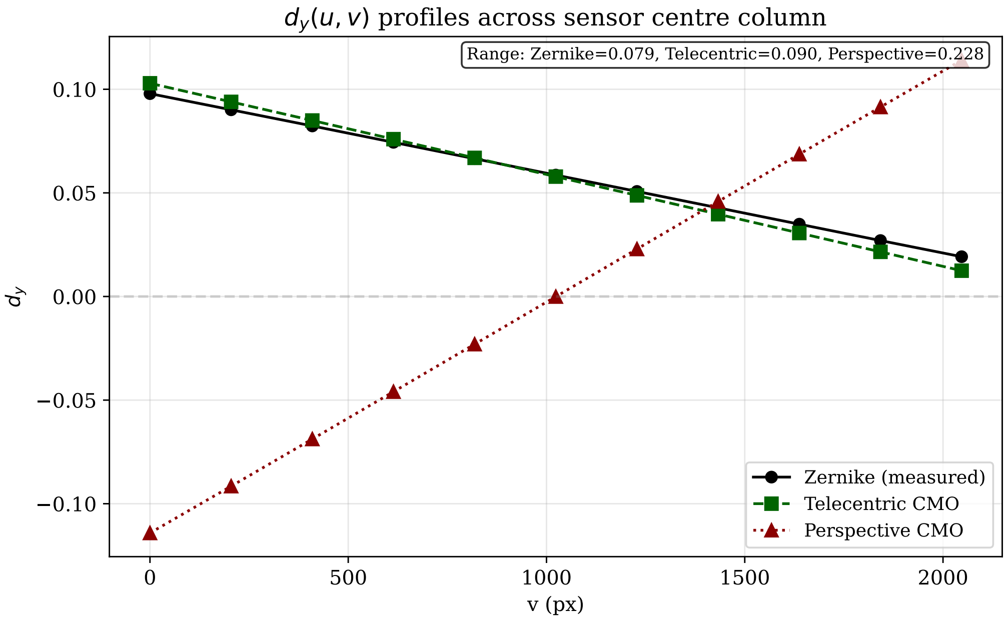

Figure 4. Comparison of $d_y(u,v)$ profiles across the sensor. The Zernike rayfield (measured) shows a nearly constant d_y (range 0.079), while the perspective CMO model predicts a 3× larger linear gradient (range 0.232). The telecentric model reproduces the measured profile almost exactly.

3.5.2 Telecentric CMO

The perspective model fails because the real microscope is object-space telecentric: chief rays in object space are nearly parallel to the optical axis rather than diverging from a single sub-pupil point. A perspective camera predicts $d_y \propto (v - c_y)$—a linear gradient from the sub-pupil—while a telecentric system emits a bundle of nearly parallel rays whose direction varies only weakly across the field, explaining the factor-3 discrepancy in Figure 4. The telecentric model directly parameterises the ray-field without deriving it from a point projection. For each channel $c \in \{L, R\}$, the origin is a rigid sub-pupil:

$$O_c = S_c = (\pm b/2,\ 0,\ WD - f_{\text{obj}}), \tag{7}$$

where the sign is + for right and − for left. The direction field is an affine function of normalised angular coordinates $\tilde{u} = (u - c_x)p/f_{\text{ang}}$, $\tilde{v} = (v - c_y)p/f_{\text{ang}}$, where p is the pixel pitch, f_{ang} is the angular focal length, and (c_x, c_y) is the principal point:

$$d_c(u,v) = \text{normalise}(d_{c,0} + s_{x,c}\,\tilde{u}\,e_x + s_{y,c}\,\tilde{v}\,e_y + s_{xy,c}\,\tilde{u}\,\tilde{v}\,e_x + s_{yx,c}\,\tilde{v}\,\tilde{u}\,e_y)\,, \tag{8}$$

where $d_{c,0} = (\pm\sin(\Theta/2),\ d_y^{\text{com}},\ \cos(\Theta/2))$ is the chief-ray direction, with Θ the full convergence angle and d_y^{com} a shared vertical component.

Adding pupil shear—an affine transverse variation of the origin—completes the model:

$$O_c(u,v) = S_c + P_{d_c}^{\perp}(\rho_{x,c}\,\tilde{u}\,e_x + \rho_{y,c}\,\tilde{v}\,e_y), \tag{9}$$

where $P_d^{\perp} v = v - (v \cdot d)\, d$ projects onto the plane orthogonal to d, ensuring $O_c \cdot d_c = 0$.

The 14 parameters are: f_{obj}, WD, b (skeleton), c_x, c_y (principal point), f_{ang} (angular scale), Θ, d_y^{com} (chief-ray direction), $s_{x,L}$, $s_{y,L}$, $s_{x,R}$, $s_{y,R}$ (per-channel slopes), ρ_x, ρ_y (shared pupil shear). This model achieves $0.12\,\text{mm}$ ray-space RMS (down from $3.48\,\text{mm}$ for perspective) and $14.6\,\text{px}$ reprojection (down from $86\,\text{px}$). The dominant geometry is captured, but $14.6\,\text{px}$ is far from the Zernike reference ($0.47\,\text{px}$).

3.5.3 Residual Analysis

The residual between the telecentric model and the Zernike rayfield is computed on a 41×41 grid: $\Delta d = d_{\text{Zernike}} - d_{\text{CMO}}$, $\Delta m = m_{\text{Zernike}} - m_{\text{CMO}}$, where $m = O \times d$ is the Plücker moment [19].

Projecting on Zernike modes up to order 4 reveals that **both Δd and Δm are 97–98 % Z_0^0 (global piston)**. These percentages are L^2 energy fractions computed over the evaluation grid: the reconstructed contribution of each mode j, $\|c_j\, B_j\|^2 = \|c_j\|^2\, \|B_j\|^2$ (coefficient norm times the mode's actual sampled column norm $\|B_j\|$), divided by the total residual energy—*not* a raw coefficient share. Because the unnormalised basis (Appendix B) is not orthonormal over the grid, these are diagnostic energy indicators rather than exact orthonormal fractions; the conclusion is robust here precisely because a single mode dominates. This is the crucial diagnostic: the residual is not a spatial field distortion (which would appear in higher Zernike modes) but a **global line-bundle offset**—the entire set of rays from each channel is slightly rotated or translated relative to the Zernike reference.

3.5.4 Testing Alternative Hypotheses

Two alternative hypotheses are tested and rejected:

- **Image-space pre-warp:** a polynomial mapping $\xi = W(u,v)$ before the direction model. Affine and quadratic warps *degrade* pixel RMS (14.6 $\to$ 16.0–16.5 px).
- **Spatially varying origin:** affine and quadratic transverse origin fields with direction fixed. Only $8\,\%$ ray RMS improvement—the residual is not spatial.

Both negative results are consistent with the Z_0^0 diagnostic (they would produce spatial, not global, changes).

3.5.5 Residual-Guided Hypothesis Construction

The central methodological contribution is that physical model construction is guided by *residual structure*, not blind fitting. At each stage, the direction residual $\Delta d = d_{\text{Zernike}} - d_{\text{model}}$ is computed on a 31×31 grid and displayed as a heatmap (Figure 5; a coarser grid than the 41×41 used for the Plücker modal decomposition of §3.5.3; the heatmap serves visual diagnosis only). The pattern of the residual guides the

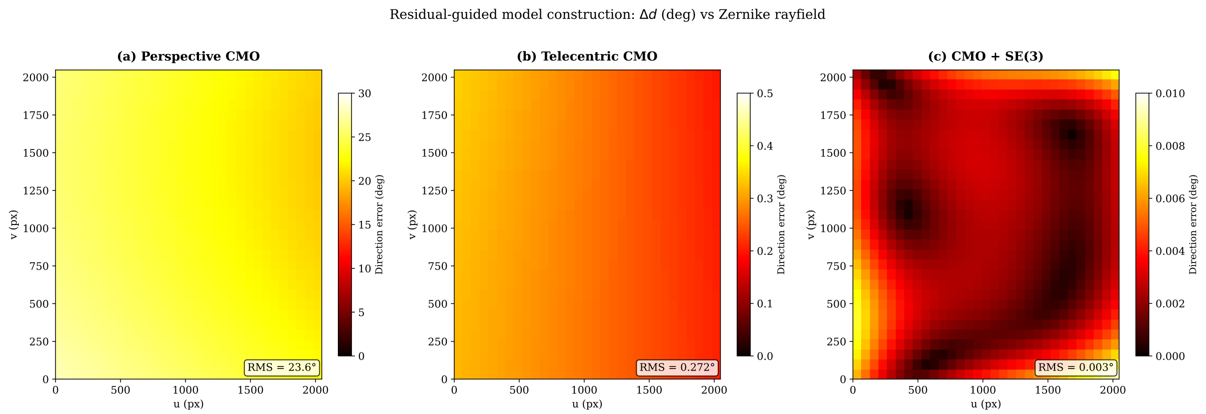

Figure 5. Residual evolution across model stages. Direction error Δd (degrees) between the Zernike rayfield (reference) and each candidate model, evaluated on a 31×31 pixel grid. **Left:** Perspective CMO — large structured error (RMS $23.6°$, max $27.7°$), dominated by a strong d_y gradient. **Centre:** Telecentric CMO — RMS drops to $0.27°$ (two orders of magnitude), but the residual is still Z_0^0-dominated (global piston), revealing the missing SE(3) arm alignment. **Right:** CMO + SE(3) — RMS drops to $0.003°$, near the noise floor. The residual heatmap at each stage guides the next modelling hypothesis, which is then confirmed or rejected by fitting the candidate model.

next physical hypothesis by indicating which class of degree of freedom is most likely missing, which is then confirmed or rejected by fitting the candidate model.

Stage 1 (Perspective CMO): the residual is large and structured, with RMS $23.6°$ (max $27.7°$). The d_y component shows a strong linear gradient trace—perspective projection from a point sub-pupil—that the Zernike field does not have. This **rejects** the perspective family and points to object-space telecentricity.

Stage 2 (Telecentric CMO): the residual drops to $0.27°$ RMS (two orders of magnitude), confirming that the telecentric model family is correct. However, the residual is Z_0^0-dominated (97–98 % global piston in both Δd and Δm)—a constant offset across the entire field. This **reveals** that the missing degree of freedom is a global line-bundle misalignment, not a spatial field distortion.

Stage 3 (CMO + SE(3)): the residual drops further to $0.003°$ RMS, near the noise floor. The remaining gap to the Zernike reference (0.47 px vs 1.06 px) comes from distributed low-amplitude aberrations (field curvature, astigmatism) that require the flexibility of the 57-parameter Zernike model.

The critical point is that **at no stage was a model added blindly**. The heatmap pattern at each stage guides the next hypothesis: perspective $\to$ telecentric (because the d_y gradient is wrong), telecentric $\to$ SE(3) (because the residual is Z_0^0 piston, not spatial).

3.5.6 Per-Channel SE(3) Arm Alignment

The Z_0^0-dominated residual points to a global misalignment of each optical arm. A rigid transform $(R_c, t_c) \in \mathrm{SE}(3)$ is applied to each channel's ray bundle, representing a small rotation and translation of the entire optical arm relative to the ideal CMO skeleton. For each pixel (u, v) on channel c, the telecentric ray $(O_c^{\mathrm{tel}}, d_c^{\mathrm{tel}})$ is transformed:

$$d_c(u, v) = R_c\, d_c^{\mathrm{tel}}(u, v), \qquad O_c(u, v) = R_c\, O_c^{\mathrm{tel}}(u, v) + t_c. \tag{10}$$

Each $R_c \in \mathrm{SO}(3)$ is parameterised by a 3-element rotation vector $r_c \in \mathbb{R}^3$ ($\theta_c = \|r_c\|$, axis $r_c/\|r_c\|$). The 12 additional parameters (r_L, t_L, r_R, t_R) bring the total to 26.

Jointly fitting the 14 telecentric parameters and the 12 SE(3) parameters against the Zernike rayfield reduces pixel RMS from $14.6\,\mathrm{px}$ to $1.06\,\mathrm{px}$ (P50=$0.87\,\mathrm{px}$, P95=$1.84\,\mathrm{px}$)—a $14\times$ improvement. The fitted rotations are $\sim 2.5°$ (left) and $\sim 3.7°$ (right); translations are sub-mm (Figure 6). Effective arm angles of this magnitude in a precision CMO instrument are best read not as a macroscopic mechanical flexure of the microscope head but as the cumulative assembly tolerances of the internal relay optics (prisms, field lenses) between the shared objective and the two sub-pupils, whose small decentrations and tilts the SE(3) term absorbs as an equivalent rigid rotation of each ray bundle. The rotation angles are stable across optimisation runs; the translation components trade off with the telecentric base parameters (a known identifiability limitation).

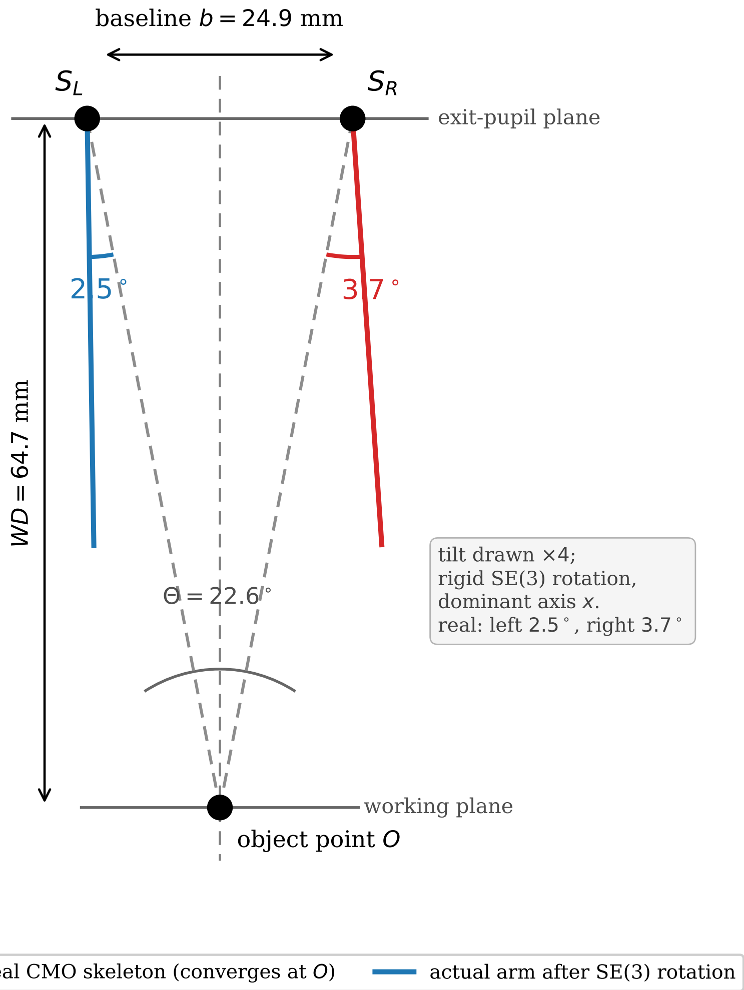

Figure 6. Per-channel SE(3) arm alignment (schematic). The ideal telecentric CMO skeleton (dashed) places two off-axis sub-pupils S_L, S_R on the baseline b, emitting chief rays that converge on the optical axis at the object point O at working distance WD with full convergence angle Θ. The fitted rigid SE(3) rotation tilts each optical arm away from this ideal by $\sim 2.5°$ (left) and $\sim 3.7°$ (right); the tilt is drawn $\times 4$ for visibility. The rotation is a rigid SE(3) of the whole ray bundle, dominated by its x-component (fitted per-axis values: left $(-2.3, -0.7, 0.6)°$, right $(-2.6, -2.7, 0.2)°$). Such effective angles arise from assembly tolerances of the internal relay optics between the shared objective and the sub-pupils, not from a macroscopic flexure of the microscope head.

3.6 Model Selection

Ray-space BIC [27] identifies the correct optical family among five candidates (pinhole, Brown-Conrady, parallel-plate, telecentric CMO with/without shear). The telecentric CMO+shear model wins by $> 40\,000$ BIC points over all alternatives.

However, this 14-parameter model produces $14.6\,\mathrm{px}$ reprojection—it is the correct *family* but not a *usable* model. We therefore complement the ray-space BIC with a **hard operational feasibility guard**; for reporting convenience the two are combined into a single score:

$$S_{\text{usable}} = \mathrm{BIC}_{\text{ray}} + \mathbf{1}_{e_{\text{px}} > 1.5\,\text{px}} \left[10^6 + N_{\text{px}} \log\left(\frac{e_{\text{px}}^2}{1.5^2} \right) \right]. \tag{11}$$

Applying this score to every candidate rejects the 14p telecentric model ($S_{\text{usable}} \approx +978\,890$) and selects the 26p SE(3)-aligned model as the best usable physical model ($S_{\text{usable}} \approx -32\,433$); the full usability-filtered comparison is reported with the results in §4.10. The 10^6 offset makes the guard effectively lexicographic: a model failing the reprojection guard can never outrank one that passes it, whatever its ray-space BIC. The selection is thus a two-stage decision—optical family by BIC, usable member by feasibility—not a weighted trade-off, which is why the 14p shear model is rejected despite having the lowest raw ray-space BIC (§4.10). The $1.5\,\mathrm{px}$ threshold is an engineering convenience, not a tuned hyperparameter: the usable model ($1.06\,\mathrm{px}$) and the rejected families ($\geq 14.6\,\mathrm{px}$ for the 14p telecentric, $> 300\,\mathrm{px}$ for OpenCV) are separated by more than an order of magnitude, so any guard in the range $\sim$2–$14\,\mathrm{px}$ yields the identical selection.

A gauge-regularisation sweep confirmed negligible Z_0-direction drift after double TPS preprocessing: even the unregularised full-pose fit already has negligible gauge drift ($0.023°$), so explicit regularisation is unnecessary.

3.7 Why Direct Bundle Adjustment is Ill-Conditioned for Non-Central Optics

The two-stage decomposition strategy (rayfield measurement $\rightarrow$ model identification) is motivated by a fundamental conditioning problem in direct bundle adjustment (BA). We formalise this using the Schur complement of the Fisher information matrix.

3.7.1 Conditioning of the Joint Estimation Problem

Direct BA jointly estimates optical parameters $\theta \in \mathbb{R}^p$ and board poses $\eta \in \mathbb{R}^q$ ($q = 6P$ for P frames) by minimising the pixel reprojection error. The Fisher information matrix at the optimum has the block structure:

$$\mathcal{I}(\theta, \eta) = \begin{bmatrix} \mathcal{I}_{\theta\theta} & \mathcal{I}_{\theta\eta} \\ \mathcal{I}_{\eta\theta} & \mathcal{I}_{\eta\eta} \end{bmatrix} = J^T J, \tag{12}$$

where $J = \partial r / \partial(\theta, \eta)$ is the residual Jacobian. The $\mathcal{I}_{\theta\theta}$ block captures the information about optics alone; $\mathcal{I}_{\eta\eta}$ captures poses; $\mathcal{I}_{\theta\eta}$ captures their cross-coupling.

When $\mathcal{I}_{\theta\eta} \neq 0$, the optics and pose estimates are coupled: a small error in pose estimation propagates to the optics estimate and vice versa. The effective information about θ after marginalising over the uncertain poses is the **Schur complement**:

$$\mathcal{I}_{\theta|\eta} = \mathcal{I}_{\theta\theta} - \mathcal{I}_{\theta\eta} \mathcal{I}_{\eta\eta}^{-1} \mathcal{I}_{\eta\theta}. \tag{13}$$

We define the **coupling norm** $c \in [0, 1]$ as the fraction of the optics information that is lost to pose coupling:

$$c = \frac{\|\mathcal{I}_{\theta\eta}\,\mathcal{I}_{\eta\eta}^{-1}\,\mathcal{I}_{\eta\theta}\|_F}{\|\mathcal{I}_{\theta\theta}\|_F}, \tag{14}$$

where $\|\cdot\|_F$ is the Frobenius norm. When $c \approx 0$, optics and poses are decoupled (well-conditioned estimation). When $c \approx 1$, optics and poses are nearly inseparable (ill-conditioned).

3.7.2 Empirical Coupling Values

The Zernike rayfield itself (57 parameters, $0.47\,\mathrm{px}$ RMS) serves as a generic non-parametric reference: it is a per-pixel spline in the Zernike basis and makes no assumptions about the optical architecture. At the rayfield-identification stage, the compact 26p model loses approximately $0.6\,\mathrm{px}$ relative to this ceiling. This $0.6\,\mathrm{px}$ gap represents, for this dataset and model family, the cost of physical interpretability: the 26 parameters capture the dominant telecentric skeleton and SE(3) arm misalignment, while the remaining $0.6\,\mathrm{px}$ come from low-amplitude aberrations (field curvature, astigmatism, detection noise) that only the flexible 57-parameter Zernike field can absorb. For applications where sub-pixel accuracy is less critical than having a physically meaningful model, this trade-off is acceptable.

In the StereoComplex direct-BA diagnostic implementation, the real Pycaso configuration yields a Schur-complement coupling norm $c = 0.98$, while a synthetic CMO-like oracle gives $c \approx 0.81$, both computed from finite-difference Jacobians (forward differences, step $\delta = 10^{-6}$, parameters scaled to comparable units). Because such block norms depend on the parameter scaling, the values are reported as conditioning diagnostics rather than as universal dimensionless properties. The qualitative interpretation is robust: c near unity indicates strong optics–pose coupling, not a universal property of all CMO microscopes. That the real instrument ($c = 0.98$) exceeds the idealised synthetic oracle ($c \approx 0.81$) is consistent with the real CMO carrying optical aberrations and mechanical misalignments absent from the synthetic model, which add to the optics–pose coupling; only the qualitative "near unity" conclusion—not the precise value—should be read into either number. A sensitivity analysis varying each parameter group's scale by ± 1 order of magnitude confirms that c is invariant under uniform rescaling of all parameters and varies by at most $\Delta c = 0.26$ under per-group perturbations (worst case: shear coefficients); across the full sweep c remains above 0.72, well within the strong-coupling regime. This means that 81–98 % of the apparent information about optical parameters is absorbed by pose coupling (depending on the diagnostic configuration)—the two parameter blocks are nearly inseparable. By contrast, in the rayfield-mediated pipeline the coupling norm $c = 0.0$ applies specifically to the pose-free ray-space model identification stage: the Zernike BA still estimates a constrained set of board poses (shared-rotation model, 15 pose parameters), but the subsequent physical model identification consumes the rayfield as a fixed observable and estimates no poses.

Figure 7 shows the normalised eigenvalue spectrum of the Schur complement S_θ on the 26-parameter CMO optical block, evaluated at the rayfield-initialised Pycaso solution. Of the 26 eigenvalues, the trailing modes (red crosses, $\lambda_i/\lambda_{\max} < 10^{-3}$) collapse after pose marginalisation—these are precisely the directions penalised by the Schur prior (Section 3.9). The coupling norm for the real Pycaso configuration is $c = 0.98$ (synthetic CMO-like oracles give $c \approx 0.81$), confirming that $98\,\%$ of the apparent optical information is absorbed by pose coupling. Schur-coupling values for the other oracles (pinhole, Brown-Conrady, parallel-plate, Greenough) were not reliably computable from the current diagnostic infrastructure and are therefore not reported.

To assess which individual parameters are identifiable after pose marginalisation, we project each of the 26 unit parameter directions onto the strong subspace (the span of eigenvectors with $\lambda_i/\lambda_{\max} \geq 10^{-3}$). The fraction of variance retained defines an **identifiability score** $\in [0, 1]$. Parameters with score > 0.5 are considered *identified* (well-observed independently of poses); parameters with score < 0.1 are *effective* (their value is driven by the gauge and regularisation, not by the data).

Only four parameters are identified: the shear coefficients s_x^L, s_y^L, s_x^R, s_y^R (scores 0.89–0.92). These encode the dominant telecentric asymmetry and are robustly determined. Two SE(3) x-rotations (rv_L^x, rv_R^x, scores ∼0.28) and the two y-translations (t_L^y, t_R^y, scores ∼0.22) are partially identifiable. All other parameters—including the effective axial length f_{obj}, the angular focal length f_{angular}, the working distance WD, the baseline b, the principal point (c_x, c_y), and the remaining SE(3) components—fall below 0.1: their individual values are effective rather than absolute, and only the linear combinations captured by the strong modes carry information independent of the pose gauge.

This confirms that the 26-parameter model provides *effective relative identification*: the compact physical skeleton (shear, telecentric structure) is data-supported, while the geometric scale (WD, b, f_{obj}) and most SE(3) components are absorbed by the rayfield gauge and pose coupling. The term "identification" in this work refers to the ray-space residual structure that selects the CMO family and its dominant asymmetries, not to the absolute metrological determination of every individual parameter.

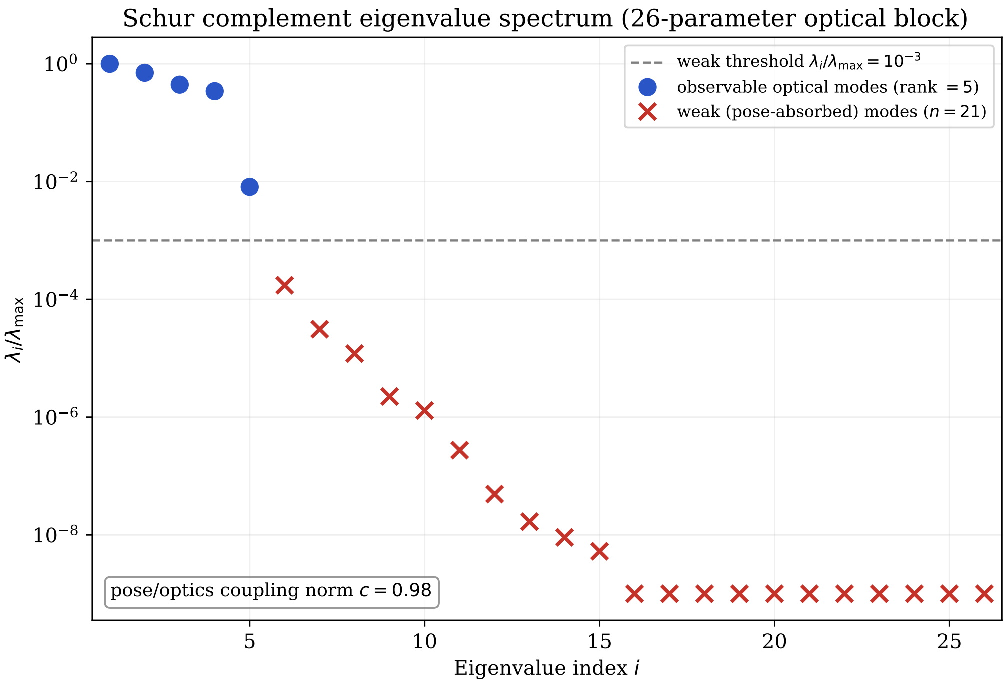

Figure 7. Normalised eigenvalue spectrum of the Schur complement S_θ on the 26-parameter CMO optical block, computed from the rayfield-initialised Pycaso model. Eigenvalues below the weak threshold $\lambda_i/\lambda_{\max} < 10^{-3}$ (red crosses) are the poorly observable directions after pose marginalisation—these are the modes penalised by the Schur prior in Section 3.9. The coupling norm for this configuration is $c = 0.98$ (synthetic CMO-like oracles give $c \approx 0.81$; see Section 3.7).

3.8 Corner Bundle Adjustment

The 26p model, fitted to the rayfield, is refined by direct corner bundle adjustment (joint optimisation of model parameters and per-frame poses on 3300 corner observations, 200 nfev, soft-L1 loss). The optimisation is performed in ray space; pixel RMS is computed afterwards by numerical inverse projection (Appendix A). The pixel RMS improves from $1.06\,\mathrm{px}$ to $0.88\,\mathrm{px}$ (P50=$0.64\,\mathrm{px}$, P95=$1.70\,\mathrm{px}$; $17\,\%$), while all 26 model parameters remain unchanged to within 10^{-3} of their rayfield-fit values. The model geometry—effective baseline, working distance, effective axial length, and convergence angle (all in the rayfield gauge and fixed f_x reference), as well as the SE(3) arm corrections—is essentially identical before and after corner refinement. This confirms that the rayfield-initialised parameters are already near-optimal for corner reprojection: the corner BA improves pose estimates but does not alter the physical model. This is a strong validation of the two-stage decomposition strategy: the rayfield fit, which never directly minimises corner error, produces parameters so close to the constrained-pose corner optimum that the dedicated constrained-pose BA can barely improve them, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement. In contrast, the direct approach—jointly estimating optics and poses from corners—is severely ill-conditioned for this non-central system (Schur-complement coupling norm $c = 0.98$ on the real Pycaso configuration, $c \approx 0.81$ on synthetic CMO-like oracles), failing to converge with a pinhole initialisation (RMS $29\,\mathrm{px}$).

3.9 Stabilising Direct BA with a Schur-Complement Prior

The corner BA of Section 3.8 improves the 2D pixel reprojection RMS by $17\,\%$ without altering the physical model. However, the Schur-complement diagnostic of Section 3.7 reveals that the Fisher information matrix at the rayfield solution θ_0 has weak eigenmodes—directions in parameter space where a small change in optics can be nearly perfectly compensated by a change in board poses. An unregularised BA can drift along these modes, trading physical interpretability for marginal reprojection gains.

3.9.1 Schur-Eigenmode Prior

The Schur complement at the rayfield solution is diagonalised,

$$S_0 = S_\theta(\theta_0, \eta_0) = V\Lambda V^T, \qquad \Lambda = \mathrm{diag}(\lambda_1, \dots, \lambda_{n_\theta}), \quad \lambda_1 \geq \cdots \geq \lambda_{n_\theta}, \tag{15}$$

where $V = [v_1, \dots, v_{n_\theta}]$ are the eigenvectors and λ_i the eigenvalues. Directions v_i with small $\lambda_i/\lambda_{\max}$ are poorly observable after pose marginalisation.

The prior penalises displacement along each eigenmode with a weight inversely proportional to its eigenvalue [32]:

$$\mathcal{L}_{\text{Schur}}(\theta) = \alpha \sum_{i=1}^{n_\theta} w_i \left(v_i^T D_\theta^{-1}(\theta - \theta_0)\right)^2, \qquad w_i = \left(\frac{\lambda_{\max}}{\lambda_i + \varepsilon\,\lambda_{\max}}\right)^p, \tag{16}$$

where $D_\theta = \mathrm{diag}(\sigma_1, \dots, \sigma_{n_\theta})$ contains per-parameter scales (degrees for rotations, millimetres for translations, pixels for the principal point), α controls the overall prior strength, p the sharpness of the eigenvalue weighting ($p = 1$ throughout), and $\varepsilon = 10^{-6}$ prevents division by zero. The prior is added to the least-squares objective as n_θ extra residual rows $r_i^{\text{prior}} = \sqrt{\alpha w_i}\, v_i^T D_\theta^{-1}(\theta - \theta_0)$.

3.9.2 Prior Sweep and Comparison with Tikhonov Baseline

Two BA configurations must be distinguished. Section 3.8 reports a *constrained-pose* corner BA: the board rotation is shared across frames and only the Z translation varies, yielding $1.06\,\mathrm{px} \to 0.88\,\mathrm{px}$ (Section 3.8) with 15 pose parameters. The sweep in this section intentionally *releases the pose constraint*: each of the 10 frames has a free 6-DOF pose (60 pose parameters), exposing the full optics–pose coupling identified by the Schur diagnostic of Section 3.7. The resulting unregularised RMS ($0.241\,\mathrm{px}$) is lower than the constrained-pose BA precisely because the free poses can absorb residual rayfield errors—which is exactly the compensation the Schur prior is designed to prevent. The two RMS values should not be conflated: the $0.88\,\mathrm{px}$ is the operational calibration result; the $0.241\,\mathrm{px}$ is a diagnostic upper bound obtained by deliberately exposing the ill-conditioned coupling.

An isotropic (Tikhonov) prior $\mathcal{L}_{\mathrm{iso}} = \alpha \| D_\theta^{-1}(\theta - \theta_0) \|^2$ serves as a baseline. Both priors are swept over $\alpha \in \{10^{-4}, 10^{-3}, 10^{-2}, 10^{-1}, 1, 10\}$ with free per-frame poses (200 nfev, soft-L1 loss). The 2D reprojection RMS is computed by numerical projection (Appendix A).

Table 2. Regularisation sweep: isotropic (Tikhonov) vs. Schur-complement prior. Δ_{weak} is the norm of the scaled parameter displacement projected onto the weak Schur eigenspace ($\lambda_i/\lambda_{\max} < 10^{-3}$). ✓/× indicates convergence.

Isotropic prior			**Schur prior**		
α	RMS (px)	Δ_{weak}	α	RMS (px)	Δ_{weak}
0	0.241 (✓)	0.934	0	0.241 (✓)	0.934
10^{-4}	0.239 (×)	0.529	10^{-4}	0.277 (✓)	**0.0033**
10^{-3}	0.245	0.248	10^{-3}	0.278	**0.0007**
10^{-2}	0.257	0.082	10^{-2}	0.279	**0.0001**
10^{-1}	0.270	0.017	10^{-1}	0.279	$< 10^{-4}$
10^{0}	0.278	0.003	10^{0}	0.283	$< 10^{-4}$
10^{1}	0.286	0.010	10^{1}	0.323	$< 10^{-4}$

Table 2 reports the results. The α=0 row is the unregularised direct BA: it converges at $0.241\,\mathrm{px}$ but at the cost of $\Delta_{\mathrm{weak}} = 0.934$, i.e. essentially *all* of the parameter drift lives in the weak Schur subspace—the optimiser exploits the poorly observable directions to chase the data residual. The isotropic prior faces a classical trade-off when trying to suppress that drift: a small α still leaves the weak modes loose ($\Delta_{\mathrm{weak}} = 0.53$ at α=10^{-4}, where the optimiser also fails to converge), while a large α degrades the fit (RMS rises to $0.286\,\mathrm{px}$). The Schur prior breaks this trade-off: even at α=10^{-4} it suppresses $99.6\,\%$ of the weak-mode drift ($0.934 \to 0.0033$, a $280\times$ reduction) while keeping the RMS within $0.038\,\mathrm{px}$ of the unregularised baseline. At α=10^{-3} the weak-mode drift is below 10^{-3} and the RMS penalty is only $0.038\,\mathrm{px}$—the prior blocks the poorly observable directions while leaving the well-observed degrees of freedom free to improve the fit. At the strongest setting (α=10), the Schur prior pays more in RMS ($0.323\,\mathrm{px}$ vs $0.286\,\mathrm{px}$ for the isotropic) but keeps the weak-mode drift below 10^{-4}, whereas the isotropic prior still leaves residual weak drift (0.010). The Schur-stabilised solution at $\alpha \leq 10^{-3}$ ($0.277\,\mathrm{px}$–$0.278\,\mathrm{px}$) is therefore not merely a diagnostic: with the optics confined to the observable subspace, the extra pose freedom improves the fit without the optics–pose compensation that makes the unregularised $0.241\,\mathrm{px}$ untrustworthy, and it is this Schur-stabilised value that we quote as the free-pose refinement of the 26-parameter model.

3.9.3 The Double Role of the Rayfield

This result formalises a double role for the rayfield estimate θ_0 in direct bundle adjustment:

1. **Initialiser.** θ_0 places the direct BA in the correct convergence basin (Section 3.8), avoiding the local minima that trap a pinhole or perspective-CMO initialisation (RMS $29\,\mathrm{px}$ vs. $0.88\,\mathrm{px}$).

2. **Observability prior.** The Schur eigenmodes of the Fisher matrix at θ_0 tell the optimiser *which directions it may trust*. The prior blocks compensation between poses and intrinsics without penalising the genuinely observable optical degrees of freedom.

The Schur prior is not a generic regulariser—it is *derived from the same data* that produced the initialisation. The rayfield's Fisher information matrix, evaluated at θ_0, encodes the local observability structure of the inverse problem. Using that structure to guide the subsequent BA is the methodological contribution: the rayfield is not merely a calibration model but an *observability oracle* for direct refinement.

4 Results

4.1 Data Sources

Table 3 clarifies which results come from real Pycaso calibration data, which from synthetic oracles, and which are methodological diagnostics.

Table 3. Data sources for results reported in this paper. Real data: 10 Pycaso stereo pairs. Synthetic oracles: StereoComplex `model_selection_oracles.py`. Method: analytical and numerical diagnostics independent of data source.

Result category	Source	Examples
Real data	Pycaso CMO images (10 stereo pairs)	Zernike fit (0.47 px), CMO+SE(3) (1.06 px), descriptors (b, WD, Θ), pose sweep, cross-validation, bootstrap, f_x sensitivity
Synthetic oracles	StereoComplex simulator (noiseless)	Direct vs. rayfield inversion ($c = 0.81$), ray-space BIC, oracle parameter recovery
Methodological	Analytical / numerical diagnostics	BIC formula, Plücker residual decomposition, SE(3) ablation, Schur derivation

4.2 Dataset and Pipeline

Ten stereo pairs of 2048×2048 px ChArUco [9, 22] images (16×12 squares, $0.3\,\mathrm{mm}$, DICT_6X6_250, `setLegacyPattern(True)`) from the Pycaso open dataset [8] are used. The Z-range is $2.65\,\mathrm{mm}$–$3.35\,\mathrm{mm}$ ($\Delta = 0.70\,\mathrm{mm}$). Corner detection yields 61–161 corners per frame; Hessian-based completion ($|\det H|$ + Otsu + barycentre) fills all 165. A double Ray2D TPS denoising pass [3] (ArUco markers at $\lambda = 10$, then completed 165 corners at $\lambda = 3$, Huber $c = 1.5$) produces gauge-stable 2D corners. The two-phase TPS is not cosmetic: without it, raw ChArUco detections carry structured biases from compression artifacts, blur, and defocus that cause the Zernike rayfield fit to become gauge-unstable—the direction piston Z_0 drifts by $8.5°$ between constrained and full-pose fits, making physical interpretation impossible. After double TPS, the gauge drift drops to $0.023°$, and the rayfield becomes a stable experimental oracle [33].

4.3 Pose Count Sensitivity

To assess robustness to the number of calibration frames, we sub-sample $N \in \{3, 5, 8, 10\}$ frames from the 10 available and re-run the full pipeline (Zernike fit + 26p model) for each. Table 4 reports pixel RMS and effective geometric descriptors.

Table 4. Pose sweep: pixel RMS and effective-descriptor stability vs. number of frames (all descriptors in the rayfield gauge with $f_x = 25600$ px). Each row is an independent refit on N frames; the N=10 value (1.07 px) reproduces the main-text calibration (1.06 px, full pipeline) to within refit noise.

N frames	RMS (px)	P50 (px)	b (mm)	WD (mm)	Θ (°)
3	1.02	0.84	24.7	64.6	22.4
5	1.03	0.84	24.8	64.6	22.5
8	1.06	0.87	24.9	64.7	22.6
10	1.07	0.88	24.8	64.7	22.4

The pixel RMS is stable within $0.05\,\mathrm{px}$ across frame counts—even 3 frames suffice for $1.02\,\mathrm{px}$ calibration. Effective descriptors vary by less than $0.2\,\mathrm{mm}$ (baseline) and $0.1\,\mathrm{mm}$ (working distance) in the chosen gauge, confirming that the pipeline is robust to the number of calibration images. The effective convergence angle $\Theta \approx 22.5°$ (standard deviation $0.1°$).

4.4 OpenCV Tuning Does Not Recover

We tested the full OpenCV stereo calibration flag space: `CALIB_RATIONAL_MODEL` (8 distortion coefficients), `CALIB_THIN_PRISM`, and `CALIB_TILTED_MODEL`. None of these configurations produced a usable calibration on the CMO data. The rational model, with the most parameters (14 intrinsics per camera), still fails because the distortion model remains radial and centred—it cannot represent a non-central projection where rays originate from two different sub-pupils. OpenCV's fisheye model (Kannala–Brandt [12]) was not pursued for the same structural reason: despite its strong radial-distortion capacity it is still a *central* projection with a single optical centre, and so cannot represent the two-sub-pupil geometry either. The best OpenCV configuration achieves $>$300 px RMS, compared to $1.06\,\mathrm{px}$ for our 26-parameter CMO+SE(3) model. This confirms that the failure is structural (central projection assumption), not a matter of under-parameterised distortion.

4.5 Geometric Descriptor Consistency

The manufacturer documentation identifies the instrument as a Leica M205C common-main-objective (CMO) stereomicroscope, equipped with a Planapo $1.0\times$ objective, an HD-V photo/video tube, a $1.0\times$ video/photo objective and a $1.0\times$ C-mount adapter. For the Planapo $1.0\times$ configuration, Leica specifies a working distance of $61.5\,\mathrm{mm}$ [13]. The datasheet does *not*, however, disclose the effective CMO stereo baseline, the sub-pupil positions, or the convergence angle—the internal geometry of the shared-objective design. These quantities are recovered here from the Zernike rayfield rather than read from the documentation, which is precisely where the rayfield is informative.

A quantitative consistency check follows from the chief-ray geometry of the telecentric CMO model used in this work. The two stereo channels view the object through two off-axis sub-pupils separated by a baseline b in the exit-pupil plane of the shared objective. By the focal (Fourier) property of the objective, a pupil point at transverse height y maps to a chief ray that leaves object space at an angle θ_y with $\tan\theta_y = y/f_{\text{obj}}$, where f_{obj} is the effective axial length (the sub-pupil-plane to working-plane distance in the CMO chief-ray model). The two sub-pupils at $\pm b/2$ therefore emit chief rays inclined at $\pm \arctan\big(b/(2f_{\text{obj}})\big)$ to the optical axis, so the full convergence angle obeys

$$\Theta = 2\arctan\frac{b}{2f_{\text{obj}}}, \qquad \text{equivalently} \qquad b = 2\, f_{\text{obj}}\, \tan\frac{\Theta}{2}, \tag{17}$$

in which the relevant axial length is the sub-pupil-plane-to-working-plane distance $f_{\text{obj}} = WD - z_p$ in the CMO chief-ray model; we do not require it to be interpreted as the manufacturer objective focal length.

The convergence angle $\Theta = \arccos(d_L \cdot d_R) = 22.6°$ is a *scale-invariant* rayfield observable—it depends only on the reconstructed ray directions, not on the chosen reference scale. Together with the reconstructed $b = 24.9\,\text{mm}$, $WD = 64.7\,\text{mm}$ and $z_p = 2.5\,\text{mm}$ (hence $f_{\text{obj}} = 62.2\,\text{mm}$), the descriptors satisfy (17) internally: $2 \times 62.2\,\text{mm} \times \tan(11.3°) = 24.9\,\text{mm} = b$, confirming that the centre-pixel rayfield obeys the CMO chief-ray geometry. The *external* check uses the only absolute length the datasheet provides. Adopting the documented working distance as the axial scale, $f_{\text{obj}} \approx WD_{\text{doc}} = 61.5\,\text{mm}$, and the measured Θ, relation (17) predicts

$$b_{\text{pred}} = 2 \times 61.5\,\text{mm} \times \tan(11.3°) = 24.6\,\text{mm}, \tag{18}$$

within $0.3\,\text{mm}$ ($\approx 1\,\%$) of the reconstructed $b = 24.9\,\text{mm}$; equivalently, the reconstructed baseline and the documented working distance imply $\Theta_{\text{pred}} = 2\arctan\big(24.9/(2 \times 61.5)\big) = 22.9°$, against the measured $22.6°$. The blind, Zernike-informed reconstruction thus recovers a CMO geometry quantitatively compatible with the documented Planapo $1.0\times$ configuration.

This agreement is a strong physical *consistency* check, not an independent metrological validation of the Leica specifications. Two caveats sharpen its scope. The internal identity (17) is close to automatic: the centre-pixel descriptors are read from the same rays, whose sub-pupil origins and directions are jointly constrained by the board at the object plane, so internal consistency mainly confirms a coherent readout. The external comparison then tests a single independent scalar—the scale-invariant Θ against the documented working distance through (17)—rather than the full descriptor set. The convergence angle is scale-invariant, but the absolute lengths (b, WD, f_{obj}) inherit the chosen reference focal length $f_x = 25\,600\,\text{px}$, and WD scales linearly with it (§4.7); reproducing the documented WD to a few percent shows only that this scale choice is compatible with the documented optics, since it still relies on the datasheet WD as the scale anchor rather than on an independent 3D measurement. What the test does establish is that the rayfield recovers internal CMO parameters absent from the manufacturer documentation, and that they are quantitatively consistent with the documented Planapo $1.0\times$ configuration. Beyond this external check, the descriptors are also **internally consistent** by three independent tests: (1) cross-validation stability: 1.07 ± 0.02 px RMS across 5 folds, with effective-descriptor spread ± 0.4 mm (baseline), ± 0.15 mm (WD); (2) bootstrap 95 % confidence intervals: $b \in [23.48, 25.04]$ mm, $WD \in [64.64, 65.13]$ mm, $\Theta \in [21.04°, 22.72°]$; (3) frame-count stability: effective descriptors vary by < 0.2 mm (baseline) and < 0.1 mm (WD) from 3 to 10 frames (Table 4).

An independent *absolute* validation—a certified 3D artefact at a known distance, or a direct measurement of the baseline or working distance—remains open and is listed among the limitations (Section 5.2). The present result nonetheless demonstrates the tool's value: from images alone, the rayfield recovers effective internal CMO descriptors absent from the manufacturer documentation and places them, through the one-scalar chief-ray check above, in quantitative agreement with the documented

Planapo $1.0\times$ configuration.

4.6 Comparison with the Soloff Polynomial Baseline

To calibrate the accuracy–interpretability trade-off, we benchmark against the Soloff polynomial model [28], a standard approach in stereo-microscope calibration [8]. A Soloff model maps 3D world coordinates (X, Y, Z) directly to pixel coordinates (u, v) via a polynomial expansion $u = \sum a_i\phi_i(X, Y, Z)$, $v = \sum b_i\phi_i(X, Y, Z)$ using monomials ϕ_i up to a chosen degree. Four independent polynomials are fitted (left-u, left-v, right-u, right-v) by ridge-regularised least squares on the same ChArUco corner observations used for our Zernike rayfield—identical input data, identical train/evaluation split. The model has no notion of 3D rays, no geometric consistency constraint between channels, and no optical interpretation of its coefficients.

Table 5 reports the pixel reprojection RMS for Soloff fits of degree 1 to 4 alongside the Zernike rayfield and the CMO 26p model.

Table 5. Pixel reprojection RMS on the Pycaso dataset: Soloff polynomial vs. Zernike rayfield vs. CMO 26p. All models are evaluated on the same 1,650 stereo corner correspondences (10 frames $\times$ 165 corners), i.e. 3,300 image observations over the two channels. Soloff fits are ridge-regularised ($\lambda = 10^{-3}$). The Zernike RMS floor ($\sim$0.41 px) is the best achievable under the 3D geometric consistency constraint; the Soloff floor ($\sim$0.18 px) is the best achievable by unconstrained 2D regression. The 0.23 px gap quantifies the cost of enforcing a physically meaningful 3D ray model.

Model	Params	RMS (px)	Interpretable
Soloff deg. 1	16	1.66 px	No
Soloff deg. 2	40	0.22 px	No
Soloff deg. 3	80	0.18 px	No
Soloff deg. 4	140	0.18 px	No
Zernike O(0)+d(2)	57	0.47 px	Rayfield
Zernike O(1)+d(3)	93	0.41 px	Rayfield
CMO telecentric 14p	14	14.6 px	Yes
CMO 26p (SE(3))	26	1.06 px	Yes

Three observations follow. First, the Soloff polynomial is a more accurate pixel regressor than any 3D-geometric model: at comparable parameter counts, Soloff deg. 2 (40 params, 0.22 px) outperforms Zernike O(0)+d(2) (57 params, 0.47 px) by a factor of $\sim$2.1$\times$, and Soloff deg. 3 (80 params, 0.18 px) outperforms Zernike O(1)+d(3) (93 params, 0.41 px) by $\sim$2.3$\times$. The polynomial reaches a noise floor of 0.18 px at degree 3, beyond which additional parameters ($\geq$ 140) produce no further improvement.

Second, the Zernike rayfield reaches its own floor at 0.41 px, approximately 0.23 px above the Soloff floor. This gap quantifies, for this dataset and model family, the cost of enforcing 3D geometric consistency: the rayfield must produce a per-pixel 3D line whose intersections with rays from the other channel are geometrically meaningful, whereas Soloff merely interpolates 2D coordinates. The gap is not a failure of the Zernike basis—it is a physical consequence of solving a harder problem.

Third, the CMO 26p model loses an additional 0.65 px relative to the Zernike reference, but gains a physically interpretable parameterisation: its 26 parameters are the baseline, working distance, effective axial length, convergence angle, shear coefficients, and per-arm SE(3) poses. No Soloff coefficient can be read as a baseline or a working distance.

These figures are the model *floors at construction*, not the endpoint. The $1.06\,\mathrm{px}$ of the CMO 26p model is its rayfield-identified *initialisation*; once that initialisation seeds the free-pose Schur-stabilised direct bundle adjustment (§3.9), the same 26 parameters refine to $0.28\,\mathrm{px}$, narrowing the gap to the Soloff degree-3 floor to under $0.1\,\mathrm{px}$. After refinement, then, the compact physical model is competitive in pixel accuracy with the black-box regressor while retaining a physically interpretable parameterisation—and it does so without the optics–pose compensation that the unregularised free-pose optimum ($0.24\,\mathrm{px}$) exploits.

The comparison thus frames the paper's central trade-off explicitly. At construction: $0.18\,\mathrm{px}$ (black-box, no geometry), $0.41\,\mathrm{px}$ (geometric rayfield, no optical model), $1.06\,\mathrm{px}$ (compact physical model, physically interpretable parameterisation). After free-pose Schur-stabilised refinement the physical model closes most of that gap, reaching $0.28\,\mathrm{px}$—so interpretability ultimately costs a fraction of a pixel rather than a category of accuracy.

This trade-off has a computational counterpart. A classical Soloff calibration is a linear least-squares fit of four polynomials and is essentially instantaneous (a few milliseconds on this dataset). The rayfield-to-CMO identification is, by contrast, an iterative non-linear optimisation: jointly fitting the Zernike rayfield with the board poses and then the 26-parameter CMO model takes $\approx$20 s on a single CPU core for the ten-pose dataset, and the optional free-pose Schur-stabilised refinement adds a comparable amount. This is a one-time, offline calibration cost—three to four orders of magnitude above a Soloff fit, but negligible in absolute terms—and once calibrated both models evaluate a pixel in closed form. The pipeline thus spends a modest offline compute budget to obtain the physical descriptors and residual diagnostics that a polynomial cannot provide.

Beyond the accuracy–interpretability trade-off, the field-resolved rayfield does what a diagnostic instrument should: it points to the next physical degree of freedom missing from the compact model. The Zernike rayfield reveals that the real instrument is not perfectly telecentric: the effective sub-pupil positions shift laterally by approximately $1.0\,\mathrm{mm}$ across the field, producing a centre-to-edge baseline variation that the 26-parameter telecentric skeleton misses entirely (it holds the sub-pupils at fixed lateral positions). This residual non-telecentricity is a natural extension of the model; but the per-parameter identifiability analysis of §3.7 shows that parameters controlling axial or lateral sub-pupil scale lie in weak directions of the planar ChArUco likelihood and cannot be assigned a metric value from image residuals alone. They require either an external reference or an explicit acquisition-metrology prior. This is consistent with the specimen comparison (§4.8), where the free-pose CMO–Zernike difference reduces to a single depth-only scale factor ($s_z \approx 0.70$), and with §4.8.1, where transmitting the nominal translation-stage ladder to the Zernike BA selects the metrically correct solution. The rayfield thus fulfills its diagnostic role not by claiming completeness, but by identifying exactly which degree of freedom remains weak—and which physical measurement can anchor it.

4.6.1 Extrapolation to Held-Out Poses

The accuracy–interpretability trade-off above is measured on the full 10-frame dataset (train = test, since both models use all available data). A stricter question is: how well does each model *extrapolate* to poses not seen during fitting? This tests whether the 3D geometric constraint in the Zernike rayfield provides a regularisation benefit beyond what the Soloff polynomial's ridge penalty achieves.

We split the 10 frames into 7 training poses (the most central ones, closest to the centroid of the pose distribution) and 3 held-out test poses (the most extreme, at the edges of the sampled volume). Both models are fitted on the training set only and evaluated on the held-out frames. Critically, the two model families have different natural output spaces—Soloff maps 3D coordinates to pixels, Zernike maps pixels to 3D rays—so their extrapolation errors are measured in their respective native metrics and compared via the *degradation ratio* (test error / train error). Because these ratios are taken in different native metrics, they are indicative rather than strictly commensurable: Table 6 reports them, and the shared-metric comparison of Table 7 below is the primary evidence.

For Soloff, the native metric is the 2D pixel prediction error (the RMS distance between the polynomial-predicted pixel and the observed corner). For the Zernike rayfield, the native metric is the 3D triangulation accuracy: observed left and right pixels are converted to rays via the rayfield, the rays are intersected to obtain a 3D point, and this point is compared to the true corner position (known from the board model and the frame pose estimated by a Kabsch alignment of the triangulated points to the board). The Kabsch step removes the rigid gauge freedom inherent to uncalibrated stereo reconstruction (the absolute pose of the board cannot be recovered without an external reference), so the residual measures only the non-rigid rayfield quality.

Table 6 reports the results. Both models degrade by a factor of approximately $1.9\times$ on held-out extreme poses, confirming that the 7-frame training set is sufficient to build a usable model in both cases. The crucial difference appears at higher model complexity: Soloff degree 3 (80 parameters) degrades by a factor of $4.65\times$ (from $0.16\,\mathrm{px}$ to $0.75\,\mathrm{px}$): its extra high-order terms lower the in-sample residual but capture training-pose-specific structure that does not transfer to the held-out extreme poses—the operational signature of overfitting. We do not claim a single mechanism; the absorbed structure may be detection noise, pose-dependent systematic residuals, or high-order instability when extrapolating beyond the training pose distribution. The Zernike rayfield (180 parameters, O(4)+d(4)), by contrast, degrades by only $1.86\times$ despite having more than twice as many parameters. The 3D geometric consistency constraint—every pixel's ray must intersect consistently with rays from the other channel and with the board geometry—acts as a structural regulariser that prevents the rayfield from memorising individual training poses.

Table 6. Extrapolation to held-out extreme poses (7 train / 3 test). Each model is evaluated in its native metric: 2D pixel error for Soloff (3D$\to$2D mapping), 3D triangulation accuracy for Zernike (2D$\to$3D mapping). The degradation ratio (test / train) is the comparable quantity across model families.

Model	Params	Train	Test (extreme)	Degradation
Soloff deg. 1	16	$1.21\,\mathrm{px}$	$2.48\,\mathrm{px}$	$2.05\times$
Soloff deg. 2	40	$0.20\,\mathrm{px}$	$0.39\,\mathrm{px}$	$1.97\times$
Soloff deg. 3	80	$0.16\,\mathrm{px}$	$0.75\,\mathrm{px}$	**4.65**$\times$
Zernike O(4)+d(4)	180	$0.0006\,\mathrm{mm}$	$0.0012\,\mathrm{mm}$	**1.86**$\times$

Two conclusions follow. First, at comparable degradation ratios ($\sim 1.9\times$), Soloff remains more accurate in its native 2D pixel metric whereas the Zernike rayfield remains stable in its native 3D triangulation metric, consistent with the in-sample comparison of Table 5: the 2D polynomial is a more accurate regressor, but its accuracy collapses when the model order is pushed (degree 3). Second, the Zernike rayfield's degradation stays bounded ($1.86\times$) even at 180 parameters because the 3D geometric consistency constraint prevents it from overfitting to individual poses. The rayfield does not learn "pixel

(500, 1300) corresponds to 3D point X in frame 3"; it learns "pixel (500, 1300) emits a ray that must intersect the corresponding ray from the other channel at a 3D point consistent with a rigid board". This constraint is absent from the per-coordinate polynomial regression, which treats each pixel coordinate as an independent scalar function of (X, Y, Z).

We deliberately do not convert the Zernike 3D error to a pixel-equivalent: that conversion factor depends on the local ray geometry (pixel pitch, working distance, ray obliquity) and the inverse map is, moreover, ill-conditioned at the microscope's magnification ($f_x = 25600$ px gives $\partial d/\partial u \approx 2.5\,\mu\mathrm{m/px}$, so a sub-millimetre ray miss displaces the inverted pixel by tens of pixels). The degradation ratio of Table 6 sidesteps this by comparing each family in its native metric.

A complementary and more direct comparison expresses *both* models in a single physical metric—the 3D reconstruction error in micrometres. This requires a shared board-pose reference: the absolute board pose is a gauge freedom that neither uncalibrated model fixes on its own, and comparing the Zernike reconstruction against the CMO-model poses alone would conflate the rayfield error with the sub-millimetre disagreement between the two bundle adjustments' pose estimates (an unaligned comparison inflates the Zernike error to $\sim$500 μm, almost entirely pose mismatch). We therefore freeze the poses and evaluate two symmetric protocols. *(i) CMO-pose reference:* the Zernike rayfield is refitted with the board poses held fixed at the CMO-model values $\{R_i, t_i\}$—only the 180 rayfield coefficients are optimised—so both models share the exact pose set that the Soloff polynomial implicitly uses; a single global rigid (Kabsch) transform then removes the residual camera-to-world gauge of the triangulated points. *(ii) Zernike-pose reference:* conversely, the per-frame poses self-consistent with the free-fit Zernike field are recovered by pose-only least squares, and the four Soloff polynomials are refitted against that geometry. Reporting both protocols brackets the reference dependence and tests whether the conclusion survives the choice of which model defines the board geometry.

Table 7. 3D reconstruction RMS (micrometres) on held-out extreme poses, with both models expressed against a shared board-pose reference. Two symmetric protocols bracket the reference dependence: poses frozen to the CMO model, or to the free-fit Zernike field. A single global rigid (Kabsch) transform, estimated on the training frames, removes the camera-to-world gauge. The qualitative ranking is invariant to the reference choice.

		CMO-pose ref.		Zernike-pose ref.	
Model	Params	Train→Test (µm)	Degr.	Train→Test (µm)	Degr.
Soloff deg. 2	40	1.24 → 2.14	1.72×	2.58 → 4.78	1.86×
Soloff deg. 3	80	0.99 → 3.81	**3.86**×	1.99 → 3.87	1.95×
Zernike O(4)+d(4)	180	1.18 → 2.39	2.04×	2.63 → 5.53	2.10×

Table 7 shows the ranking is invariant to the reference choice. In both protocols the 180-parameter Zernike rayfield *matches the accuracy of the 40-parameter Soloff degree-2 fit* (1.2 μm/2.4 μm train/test under the CMO reference, 2.6 μm/5.5 μm under the Zernike reference) and degrades by only $\sim 2\times$, never collapsing. Soloff degree 3 attains the lowest training error in both references, but its degradation is reference-dependent: against the cleaner CMO geometry (the CMO model fits all ten frames jointly) it overfits sharply ($3.86\times$), consistent with its native pixel-metric collapse ($4.65\times$, Table 6); against the noisier Zernike-recovered geometry the same overfit is partly masked ($1.95\times$) because the $\sim 2\,\mu$m pose-recovery floor inflates the training residual. The absolute micrometre values and the degree-3 overfit magnitude therefore depend on which model's poses define the reference—neither constitutes an external metrological truth, consistent with the absolute-scale caveats above—but the qualitative conclusion is robust across both: the Zernike rayfield buys 3D geometric consistency and overfit resistance at the cost of $\sim 0.2\,\text{px}$, whereas a high-order Soloff polynomial maximises raw pixel accuracy at the risk of overfitting.

4.7 Internal Validation

Three internal validation experiments assess the robustness of the 26p pipeline on the Pycaso dataset.

Cross-validation. 5-fold leave-2-out cross-validation (fit on 8 frames, evaluate on 2 held-out frames) yields a mean pixel RMS of 1.07 ± 0.02 px. The low standard deviation ($\sim 0.02\,\text{px}$) confirms that the model is not overfitting to specific frames; the calibration is stable across the 10-frame dataset. Fold 1 shows slightly higher RMS ($1.09\,\text{px}$) and a lower effective baseline estimate ($b = 23.1\,\text{mm}$) because the held-out frames (0 and 1) cover the most extreme Z values.

Bootstrap descriptor stability. 100 bootstrap iterations (resampling 10 frames with replacement, refitting the full pipeline each time) give 95 % confidence intervals:

- Effective baseline b: $[23.48, 25.04]$ mm (mean 24.56 mm, $\sigma = 0.43$ mm)
- Effective working distance WD: $[64.64, 65.13]$ mm (mean 64.86 mm, $\sigma = 0.15$ mm)
- Effective convergence angle Θ: $[21.04°, 22.72°]$ (mean 22.17°, $\sigma = 0.46°$)

The effective working distance is the most stable descriptor ($\sigma < 0.2\,\text{mm}$), while the effective baseline and convergence angle show $0.4\,\text{mm}$ and $0.4°$ variation respectively—consistent with the gauge sensitivity of the Zernike origin field noted in Section 5.2.

Sensitivity to fixed focal length f_x. Varying the fixed pinhole focal length f_x by $\pm 10\%$ (5 values: 23040, 24320, 25600, 26880, 28160 px) produces:

- Pixel RMS: $1.21 \to 0.95$ px—remains near the one-pixel range, confirming that the rayfield is stable as a calibration object.
- Working distance: $58.3 \to 71.3$ mm—linearly proportional to the initial fx ($\pm 10\%$ fx $\to \pm 10\%$ WD). WD is directly scaled by the pinhole reference and is *not* independently identifiable.
- Convergence angle: $24.1° \to 20.5°$—varies inversely with WD.
- Baseline: $23.9 \to 25.1$ mm ($\pm 2.5\%$)—moderately stable.

The absolute physical descriptors, especially WD and θ, remain sensitive to the chosen pinhole reference scale. Descriptor readout therefore requires either an independently known optical scale (pixel pitch, magnification) or external metrological validation—a known limitation of all single-view calibration methods without a known reference object.

4.8 Application Sanity Check: Specimen Reconstruction

A non-metrological sanity check validates the calibrated rayfields on an independent speckled coin specimen imaged with the same Pycaso microscope (single stereo pair, no ChArUco pattern). This is **not** absolute metrological certification against a traceable 3D artefact, but—together with the external profilometry of the same specimen (§4.9)—it confirms that the rayfields produce coherent dense reconstruction on data unseen during calibration. Dense left-to-right pixel correspondences are computed by DIS optical flow on the *full* image (with variational refinement), and the analysis ROI is then extracted from the resulting field rather than computing the flow on a cropped ROI; within the 1448×1448 px analysis window shown in Figure 8, every pixel carries a valid correspondence (the full-image flow leaves no holes in the ROI). For each correspondence, the two rays from the calibrated model are intersected at their closest point (ray-gap triangulation).

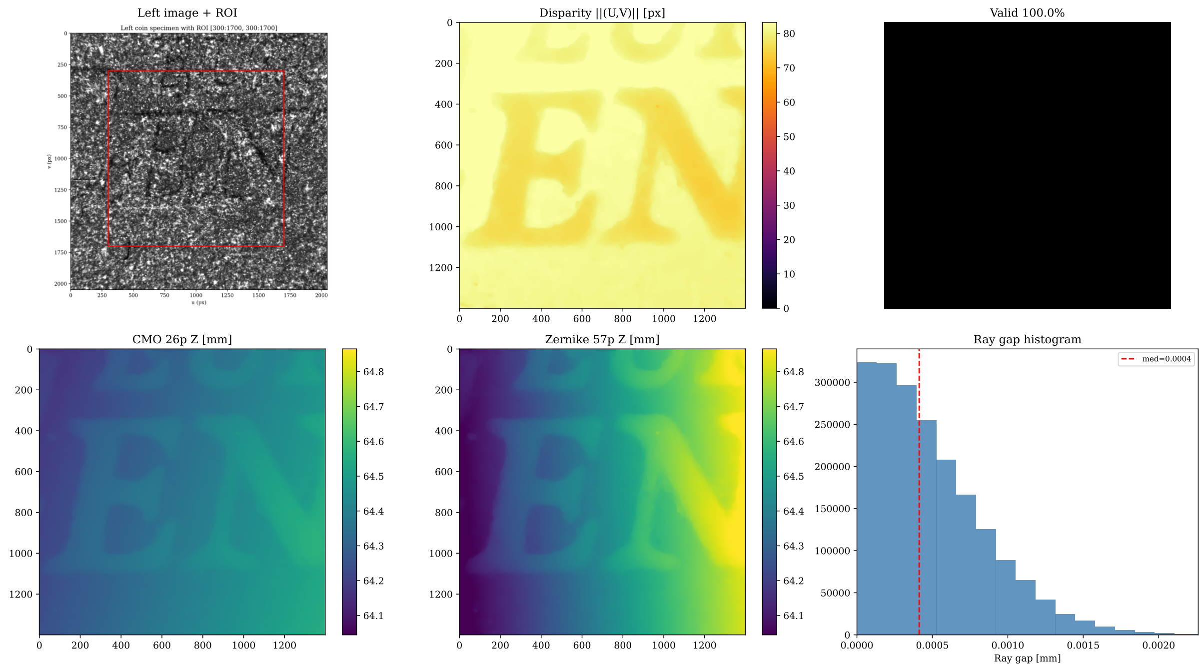

Figure 8. Dense specimen reconstruction. Top row: left image, DIS optical-flow magnitude $\|(U, V)\|$, valid mask (full coverage—the full-image flow leaves no holes in the ROI). Bottom row: reconstructed Z (depth) maps from the CMO 26p and the field-resolved Zernike O(2)+d(2) rayfield (87 parameters including the 15 constrained-pose parameters), and the ray-gap histogram. Both rayfields produce nearly identical, tight ray gaps (median $\sim 0.5\,\mu\text{m}$), confirming rayfield coherence. Quantitative surface roughness and ray-gap values for all five model variants (rayfield-initialised through Schur-regularised) are reported in Table 8.

The median ray gap is now $0.4\,\mu\text{m}$ to $0.5\,\mu\text{m}$ across the displayed CMO and Zernike reconstructions in Figure 8, and all five variants in Table 8 show similarly tight intersections, confirming that the two channels' rays intersect each other at the sub-micrometre level despite never having been jointly optimised on this specimen. The CMO 26p reconstructions are smoother than the Zernike rayfield reconstruction because the compact physical model imposes implicit spatial smoothness through its rigid-sub-pupil and affine-direction parameterisation, whereas the 72-coefficient field-resolved Zernike model has the flexibility to absorb DIS correspondence noise into high-frequency surface variations. Quantitative Z MAD and ray-gap values for the rayfield-initialised, unregularised BA, isotropic-prior BA, and Schur-prior BA variants are reported in Table 8. OpenCV stereo reconstruction was not available because the standard OpenCV calibration failed to converge on this microscope's calibration data.

The specimen comparison is intentionally presented in two steps, because the apparent disagreement is scientifically useful. First, the SE(3) alignment shows that both models reconstruct the same surface up to the CMO model's $v \to Y$ sign convention (a Y-axis reflection) and a small ($\sim 8.6°$) rigid rotation. Second, after these gauge effects are removed, a residual depth-relief discrepancy remains: the Zernike reconstruction exhibits approximately $42\,\%$ larger relief amplitude than the CMO 26p model (standard deviation ratio $1.42\times$, Figure 9). To characterise whether the discrepancy is a uniform scale error, a

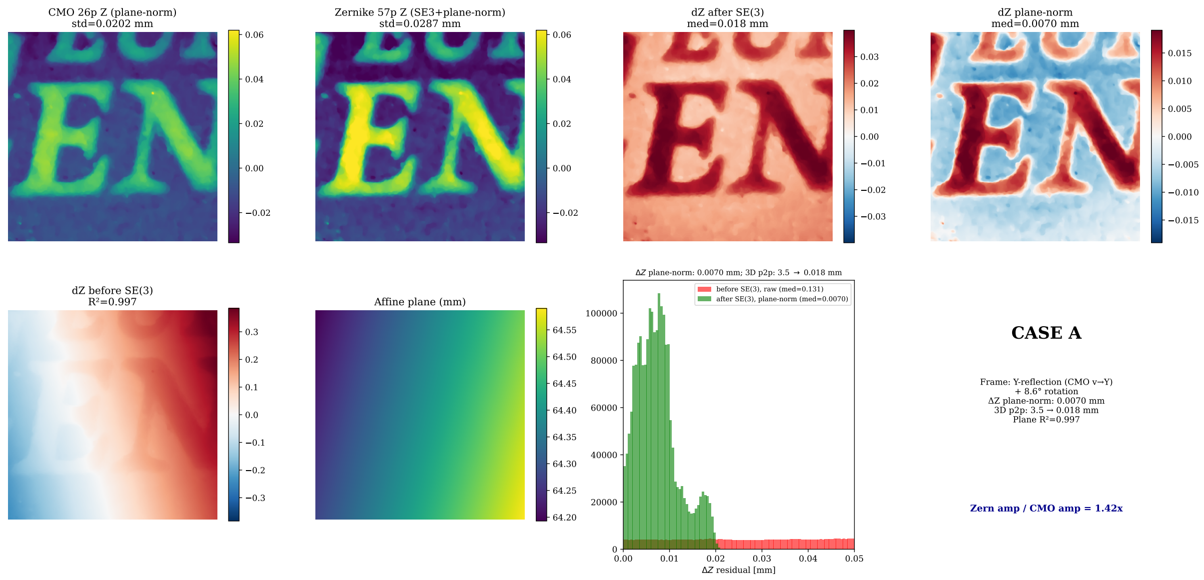

Figure 9. Rigid-gauge removal: Zernike vs CMO surface comparison. Top row: CMO and Zernike surfaces after plane-normalisation (relief only), followed by ΔZ before and after plane normalisation. Bottom row: ΔZ before SE(3) alignment (dominated by an affine ramp, $R^2 = 0.997$), the fitted affine plane, and the residual histogram. After SE(3) alignment the median plane-normalised ΔZ residual is $0.0070\,\mathrm{mm}$; the corresponding median 3D point-to-point residual drops from $3.5\,\mathrm{mm}$ to $0.018\,\mathrm{mm}$ (a $99.5\,\%$ reduction). The remaining relief difference is a depth-only scalar (relief-amplitude ratio $Z_{\mathrm{Zernike}}/Z_{\mathrm{CMO}} = 1.42\times$, i.e. $s_z \approx 0.70$): the two reconstructions agree up to a single depth-scale factor, not a spatially arbitrary non-rigid discrepancy. This diagnosis motivates the stage-metric observability prior tested immediately below and independently checked by profilometry in §4.9.

non-rigid shape error, or a depth-only effect, we fit nested transformations from Zernike to CMO points on matched specimen correspondences. An SE(3)-only fit (no scale) yields a median residual of $0.008\,\mathrm{mm}$, already confirming excellent geometric agreement. This $0.008\,\mathrm{mm}$ is the SE(3)-only fit residual on the 50 000 subsampled correspondences, distinct from the $0.018\,\mathrm{mm}$ dense-map median 3D point-to-point residual reported in Figure 9. Adding a global isotropic scale (Sim(3)) produces an optimal scale factor $s = 1.00$, ruling out a uniform 3D magnification—the difference is not a simple focal-length or pixel-scale error. However, an anisotropic scale model with independent (s_x, s_y, s_z) recovers $s_x \approx s_y \approx 1.00$ but $s_z \approx 0.70$, indicating a **depth-only scale difference** of approximately $30\,\%$ (consistent with the $1.42\times$ relief-amplitude ratio above). The lateral dimensions (x, y) are essentially identical between the two reconstructions; only the reconstructed depth range differs, by the scalar factor $s_z \approx 0.70$ above.

This depth-only scale difference is therefore not a non-rigid disagreement and not a contradiction of the CMO family identification: the two reconstructions agree up to a single depth-scale factor. The CMO model's rigid sub-pupil and telecentric direction parameterisation impose a specific stereo geometry that the Zernike rayfield, being free, is not obliged to reproduce exactly.

4.8.1 Stage-Metric Observability Prior

The residual axial freedom is a property of the calibration pose hypothesis, not an intrinsic limitation of the Zernike basis. The Pycaso frames are near-parallel planes translated by a motorised stage: they constrain lateral geometry strongly, but a change in axial ray gain can be compensated by changing the recovered inter-plane distances. In the baseline BA each $t_{z,i}$ is free, so the optimiser is asked to infer both the rayfield and the metric ladder from the same nearly fronto-parallel observations. Equation (5) instead supplies the acquisition metrology as an observability prior while retaining an estimated stage axis, one common scale, and frame-specific positioning errors.

This construction puts the Zernike and Soloff calibrations on the same metrological footing. The standard Pycaso Soloff workflow first assigns every calibration plane the axial coordinate stored in the image filename, `z_list`, and fits the polynomial against those 3D coordinates. The direct fit therefore corresponds to the hard-ladder limit of Equation (6). The optional Soloff retroprojection iteration updates the individual plane coordinates, but an unobservable common rescaling has no data gradient; its global metric is consequently inherited from the nominal initial ladder. The closest hierarchical analogue is a tightly anchored s with looser ε_i. The two algorithms are not algebraically identical, but they receive the same physical information: the nominal stage ladder supplies the common axial metric, while image consistency may correct relative plane placement.

We test four pose hypotheses on the field-resolved O(2)+d(2) rayfield used for the specimen reconstruction (72 rayfield coefficients plus 15 baseline pose parameters, 87 total), using the same 1,650 stereo corners, fixed reference intrinsics, dense DIS correspondence field, and profilometry registration throughout:

1. *free axial poses*: the published baseline, with no stage residual;
2. *weak hierarchy*: $\sigma_s = 10\%$, $\sigma_\varepsilon = 70\,\mu\mathrm{m}$;
3. *strong hierarchy*: $\sigma_s = 0.1\%$, $\sigma_\varepsilon = 5\,\mu\mathrm{m}$;
4. *near-fixed ladder*: $\sigma_s = 0.1\%$, $\sigma_\varepsilon = 1\,\mu\mathrm{m}$.

The ray residual scale is fixed to $\sigma_r = 0.58\,\mu\mathrm{m}$, measured from the unregularised fit; profilometry is used only for evaluation, never in the BA objective.

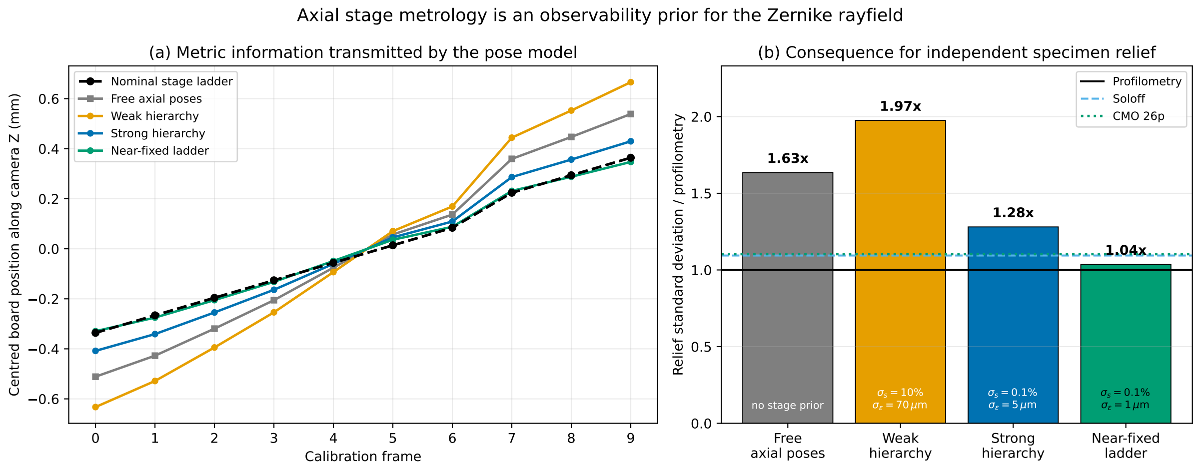

Figure 10. Translation-stage metrology as an observability prior for the Zernike rayfield. (a) Recovered board positions after removing their common translation gauge. The free and weakly regularised pose models stretch the nominal $0.70\,\mathrm{mm}$ ladder; tightening the common scale alone is insufficient because the frame offsets ε_i can still absorb axial gain, whereas the near-fixed ladder follows the nominal positions. (b) Independent specimen relief amplitude, normalised by the profilometric standard deviation. The free rayfield over-amplifies the relief ($1.63\times$), and the weak hierarchy worsens it ($1.97\times$). A strong scale prior with moderate jitter reduces the gain to $1.28\times$; only the near-fixed ladder reaches $1.04\times$, consistent with the Soloff ($1.09\times$) and CMO 26p ($1.10\times$) references. All three hierarchical fits retain essentially identical ray-space RMS ($0.582\,\mu\mathrm{m}$ to $0.588\,\mu\mathrm{m}$), showing that the calibration likelihood alone cannot select the metrically correct member of this weak family.

Figure 10 changes the interpretation of the discrepancy. The common scale is not an isolated one-dimensional null mode once high-order origin coefficients, the estimated stage axis, and per-frame offsets are fitted jointly: s, ε_i, and the rayfield coefficients form a broader weak subspace. Consequently, the nominally "weak" prior does not automatically determine only an otherwise flat direction. It converges to $s = 1.456$ with $105\,\mu\text{m}$ RMS frame offsets, expands the recovered ladder to $1.299\,\text{mm}$, and increases the specimen gain to $1.97\times$ while barely changing the corner likelihood. The weak hierarchy is intentionally permissive: with $\sigma_\varepsilon = 70\,\mu\text{m}$, the allowed frame-to-frame jitter is of the same order as the nominal stage increment ($\sim$70 μm). It therefore does not merely regularise a single axial scale mode, but leaves a broader weak subspace in which the global scale s, the individual ε_i, and the rayfield coefficients can compensate each other. This explains why the weak hierarchy can yield a larger dense-depth gain than the free-pose solution. Even fixing the common scale to $s = 1.005$ leaves enough per-frame freedom to produce a $0.838\,\text{mm}$ ladder and $1.28\times$ relief.

By contrast, the near-fixed ladder recovers $s = 0.998$, a $0.677\,\text{mm}$ axial span, and a specimen relief of $1.04\times$ the profilometry, without degrading the ray-space fit. The near-fixed ladder should therefore be read as a **nominal-ladder, Soloff-equivalent limit**: it transmits to the Zernike rayfield the same nominal axial plane coordinates that are already supplied to the direct Soloff calibration through the `z_list`. It is not used here as an independent claim that the absolute stage calibration is known to $0.1\,\%$ or $1\,\mu\text{m}$. The stage manufacturer's specified absolute accuracy is approximately $10\,\%$ of the nominal step; the prior hyperparameters σ_s and σ_ε control the relative weighting of scale uncertainty and per-frame jitter within the hierarchical model, not the absolute metrological error of the translation stage. The profilometry data are not used to tune these hyperparameters; they serve only as an external check, applied after the BA, of the dense reconstruction selected by each pose-metrology assumption. Thus the original $1.63\times$ result is not an unavoidable over-amplification of a high-dimensional non-central rayfield. It is the metrically unconstrained solution selected when the known stage ladder is discarded from the BA. Transferring essentially the same axial-coordinate prior used by direct Soloff calibration restores the Zernike depth gain; the external profilometry then serves as an independent validation that the stage-anchored member of the weak solution family is the physically appropriate one for this specimen.

Table 8 quantifies the five reconstructions. All five variants—including the rayfield initialisation—recover tight ray intersections on the dense full-image correspondence field, with a median ray gap of $\sim$0.5 μm. The ray geometry is therefore correct regardless of the regularisation strategy, and the rayfield-identified initialisation already triangulates the specimen as tightly as the BA-refined models. The regularised variants achieve Z MAD $\sim$0.028 mm, comparable to the unregularised BA ($0.023\,\text{mm}$)—the prior keeps the dense reconstruction comparable while preserving ray-gap and magnification consistency. On the dense full-image correspondence field all five variants converge to the same lateral magnification ($\sim$0.199 vs. the nominal coin): there is *no* systematic CMO–Zernike lateral bias. The apparent $\sim$3.5 % bias seen with the sparser ULTRAFAST correspondences was an artefact of differing per-variant valid sets, not a property of the models. The lateral magnification, now consistent across variants, is in any case distinct from the free-pose Zernike depth-gain difference ($s_z{\approx}0.70$) and cannot explain it: the specimen diagnostic separates lateral magnification from depth sensitivity, while Figure 10 shows that the latter is controlled by the axial pose prior.

Table 8. Five-variant specimen reconstruction: surface roughness (Z MAD), median ray gap, and magnification ratio vs. the nominal 18.75 mm 2-cent euro coin.

Model	Z MAD (mm)	Median gap (mm)	Magnification
Zernike rayfield O(2)+d(2) (87 p incl. poses)	0.202	0.0005	0.1991
CMO 26 p (rayfield init)	0.076	0.0004	0.1990
CMO 26 p — BA unregularised	0.023	0.0005	0.1979
CMO 26 p — BA + isotropic prior (α=10^{-2})	0.028	0.0004	0.1985
CMO 26 p — BA + **Schur prior** (α=10^{-3})	**0.028**	0.0004	0.1985

4.9 External Profilometry Validation

With free per-frame axial poses, the stereo likelihood cannot determine which member of the weak depth-gain family is metrically correct. Section 4.8.1 selects an internal metric candidate by transferring the acquisition-stage ladder into the BA; the independent question is whether that candidate agrees with the real specimen. We answer it with a profilometric scan of the same coin. The scan covers the embossed letter "E" and is registered to the left-camera pixel grid by a rigid similarity (scale $\approx$1.03) fitted on the binary letter mask (intersection-over-union 0.88), a flow-independent alignment that fixes the "E" geometry. Dense correspondences are recomputed with a refined DIS flow (full image, variational refinement), and the *same* field is reconstructed by Soloff, the CMO 26p model, and the near-fixed-ladder Zernike rayfield selected in Figure 10. Each plane-normalised relief is compared, on the common "E" window, against the profilometry.

The result (Figure 11) resolves the ambiguity for this specimen and ROI. Against the profilometric relief standard deviation of $20.8\,\mu\text{m}$, the Soloff polynomial recovers $22.8\,\mu\text{m}$ ($1.09\times$), the CMO 26p model $23.0\,\mu\text{m}$ ($1.10\times$), and the stage-anchored Zernike rayfield $21.6\,\mu\text{m}$ ($1.04\times$); all three correlate with the ground-truth shape at $\text{cc} = 0.958$. The CMO 26p and Soloff reconstructions remain statistically *indistinguishable* on the relief (mutual $\text{cc} = 0.9997$, RMS difference $0.57\,\mu\text{m}$). The stage-anchored Zernike map is likewise nearly identical to Soloff (mutual $\text{cc} > 0.99999$, RMS difference $1.21\,\mu\text{m}$) while its amplitude is the closest of the three to the profilometry.

Two conclusions follow. First, after putting all three calibration families on the same stage-metric footing, their relief amplitudes agree with the external profilometry within $4\,\%$ to $10\,\%$; the Zernike rayfield no longer forms an outlier. The remaining differences are consistent with dense-stereo correspondence noise and with the imperfect lateral registration itself (mask $\text{IoU} = 0.88$, point-wise $\text{cc} = 0.958$, not unity), and should not be read as calibrated metrological uncertainties. Second, Figures 10 and 11 separate diagnosis from validation: the former shows that the free-pose likelihood admits different axial gains and that the stage ladder selects the $1.04\times$ solution; the latter shows spatially, over the complete "E" relief and an independent depth profile, that this selected Zernike solution agrees with the specimen. The original $1.63\times$ result therefore identifies a missing *pose-metrology prior*, not an intrinsic limitation of Zernike or non-central rayfields.

The production DIS parameters were selected after inspecting the profilometric agreement; the comparison is therefore an external *consistency check of the calibrated models on a fixed dense correspondence field*, not a blind validation of the DIS front-end. To confirm that this tuning does not manufacture the free-pose discrepancy, Table 9 repeats the comparison under the bare OpenCV ULTRAFAST / FAST / MEDIUM preset defaults (no manual variational refinement): across every setting

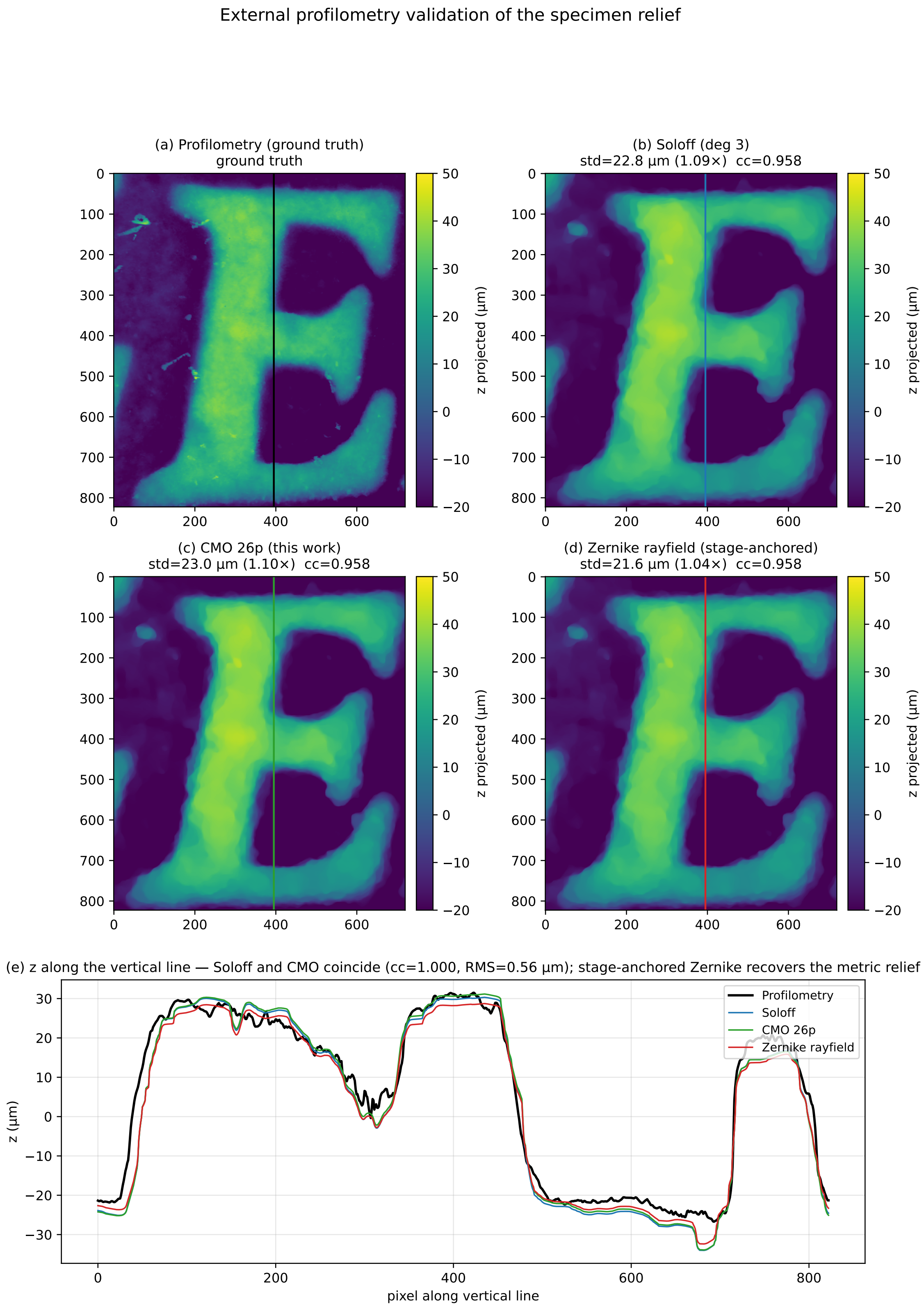

Figure 11. External profilometry validation of the specimen relief after applying the pose prior selected in Figure 10. Maps (a–d): plane-normalised relief of the embossed "E" from the profilometric ground truth, the Soloff polynomial, the CMO 26p model, and the near-fixed-ladder Zernike rayfield, on a common registered window. The annotated amplitude ratios are $1.09\times$, $1.10\times$, and $1.04\times$, respectively, and all three stereo reconstructions track the profilometric shape with $\mathrm{cc} \approx 0.958$. Panel (e): depth profiles along the same vertical line. Soloff and CMO nearly coincide; the stage-anchored Zernike profile follows the same plateaus and valleys without the axial over-amplification of the free-pose solution shown in Figure 10.

Table 9. Robustness of the relief comparison to the dense-matching front-end. Amplitude ratios are the plane-normalised relief standard deviation relative to the profilometry; the last two columns quantify the CMO–Soloff agreement. The production DIS parameters were tuned against the profilometry, but the bare OpenCV preset defaults (no manual variational refinement) reproduce the same relative result: CMO ≈ Soloff and the Zernike ray-field over-amplifies. The external comparison is therefore a consistency check of the calibrated models on a fixed correspondence field, not a tuning of the DIS front-end.

DIS setting	Soloff ($\times$)	CMO 26p ($\times$)	Zernike ($\times$)	CMO–Soloff RMS (μm)	CMO–Soloff cc
Tuned (production)	1.09	1.10	1.63	0.56	0.9997
ULTRAFAST (defaults)	1.09	1.10	1.63	0.58	0.9997
FAST (defaults)	1.06	1.07	1.59	0.56	0.9997
MEDIUM (defaults)	1.12	1.12	1.67	0.58	0.9997

the CMO and Soloff reliefs stay indistinguishable (CMO–Soloff RMS $0.56\,\mu\text{m}$ to $0.58\,\mu\text{m}$, $\text{cc} = 0.9997$) and the free-pose Zernike rayfield keeps over-amplifying (1.59–$1.67\times$). The diagnosis of a pose-metric gain is therefore invariant to the dense-matching front-end; the stage-prior experiment then changes that gain while keeping the tuned correspondence field fixed.

The Soloff polynomial is therefore used here as a strong black-box metrological baseline, not as an optical diagnostic. Even when it reconstructs the coin relief accurately, its polynomial coefficients do not define per-pixel rays, Plücker residuals, baselines, convergence angles, or Schur-observable optical modes. It validates surface reconstruction but cannot replace the rayfield-to-CMO identification step.

4.10 Model Performance

Table 10. Model comparison on Pycaso CMO data. The 26-parameter CMO+SE(3) model is the compact physical reference. All rows use the operational constrained-pose protocol; the free-pose Schur-stabilised refinement of the same 26-parameter model reaches $0.28\,\text{px}$ (Table 2) and is reported separately because its pose model (free 6-DOF per frame) differs from the constrained protocol of this table.

Model	p	Ray RMS (mm)	Px RMS (px)	P50 (px)	P95 (px)
OpenCV stereo	–	–	> 300	–	–
Perspective CMO	19	3.48	$\sim$86	–	–
Telecentric CMO + shear	14	0.12	14.6	13.2	22.4
CMO + SE(3)	**26**	**0.0021**	**1.06**	**0.87**	**1.84**
CMO + SE(3) + corner BA	26	$\sim$0.0019	0.88	0.64	1.70
Zernike O(0)+d(2)	57	0.0007	0.47	0.34	0.86

Table 10 compares all model families on the Pycaso data. The 26-parameter CMO+SE(3) model achieves $1.06\,\text{px}$ reprojection with less than half the parameters of the Zernike reference. It has a physically interpretable parameterisation: sub-pupils, effective axial length, telecentric slopes, and SE(3) arm corrections.

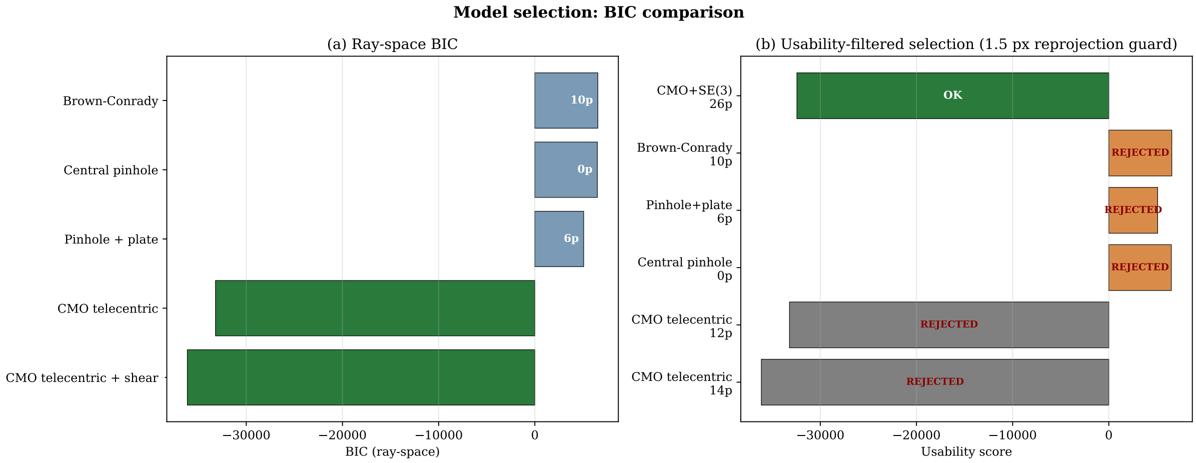

Figure 12. Two-panel BIC bar chart. (a) Ray-space BIC: the CMO telecentric family achieves the lowest BIC by a decisive margin ($> 40\,000$ points). (b) Operational usability score with a $1.5\,\mathrm{px}$ reprojection guard: models exceeding the threshold are marked as REJECTED (shown in grey/orange); only the 26p SE(3)-aligned model passes the guard (shown in green).

4.11 BIC Model Selection

Table 11. BIC model selection on the Zernike rayfield. Models above 1.5 px reprojection are REJECTED by the operational guard. The central and parallel-plate models have multi-millimetre ray RMS and are rejected on reprojection alone; their observability-penalised score S_{usable} then diverges ($\gg 0$) and is not tabulated precisely.

Model	p	RMS (mm)	BIC$_{\mathrm{ray}}$	S_{usable}	Status
cmo telecentric shear	14	0.111	-36 128	+978 890	REJECTED
cmo telecentric	12	0.146	-33 201	+986 044	REJECTED
pinhole parallel plate	6	6.337	5 079	$\gg 0$	REJECTED
central pinhole	0	8.343	6 500	$\gg 0$	REJECTED
central brown conrady	10	8.324	6 549	$\gg 0$	REJECTED
CMO + SE(3) 26p	26	0.002	$-32\,433$	$-32\,433$	**BEST USABLE**

Table 11 and Figure 12 report the usability-filtered selection. The ray-space BIC provides decisive rejection of the non-CMO families (ΔBIC $> 40\,000$), confirming that pinhole, Brown-Conrady, and parallel-plate models are grossly misspecified for this non-central system. The BIC magnitudes should be interpreted as rejection of families, not as certification of the retained model: the rayfield residuals are spatially correlated (the Zernike field is smooth), so the effective number of independent observations is lower than $N_{\mathrm{obs}} = 3300$, inflating the absolute BIC scale. The ranking across families is robust to this effect. The operational usability score correctly rejects the 14p base model as unusable and selects the 26p aligned model.

4.12 Geometric Descriptors

Table 12 collects the CMO-consistent effective geometric descriptors read directly from the centre-pixel rayfield under the chosen gauge.

Table 12. CMO-consistent effective geometric descriptors read directly from the Zernike rayfield at the centre pixel. All values are reported in the chosen rayfield gauge ($O \cdot d = 0$) and fixed reference focal scale ($f_x = 25600$ px); their absolute metrological interpretation requires an independent external reference (see Section 5.2).

Descriptor	Symbol	Value
Stereo baseline	b	$24.9\,\mathrm{mm}$
Sub-pupil depth	z_p	$2.5\,\mathrm{mm}$
Working distance	WD	$64.7\,\mathrm{mm}$
Effective axial length	f_{obj}	$62.2\,\mathrm{mm}$
Convergence angle	Θ	$22.6°$

4.13 SE(3) Ablation

Table 13. SE(3) arm alignment ablation. Per-arm DOFs are individually necessary.

Variant	Params	Ray RMS (mm)	Px RMS (px)	P50 (px)	P95 (px)
Telecentric baseline	14	0.12	14.6	13.2	22.4
+ Rotation only	20	0.014	3.74	2.68	7.13
+ Translation only	20	0.010	2.44	2.06	4.15
+ Full SE(3)	**26**	**0.0021**	**1.06**	**0.87**	**1.84**
+ Shared rotation	23	0.0083	–	–	–
+ Shared translation	23	0.0041	–	–	–
+ Differential only	20	0.0070	4.04	2.85	7.74

Both rotation and translation are essential; neither alone suffices (Table 13). Per-arm DOFs are individually necessary—any compression degrades the fit ($+92\,\%$ to $+289\,\%$ ray RMS increase).

5 Discussion

5.1 The Rayfield as Diagnostic Instrument

The key methodological contribution is the feedback loop: Ray2D preprocessing $\rightarrow$ Ray3D measurement $\rightarrow$ residual modal analysis $\rightarrow$ hypothesis $\rightarrow$ model refinement $\rightarrow$ verification. Without the rayfield, the 2D reprojection error tells you *that* the model is wrong but not *how*. The Zernike projection of Δd and Δm directly identifies whether the missing DOF is global (low-order Zernike modes) or spatial (high-order modes).

Figure 5 illustrates this diagnostic value concretely. The residual heatmap at each model stage is a **visual hypothesis generator**: the perspective CMO's structured d_y error (23.6° RMS) sends the user to telecentric models; the telecentric model's uniform Z_0^0 piston ($0.27°$ RMS) sends the user to global arm alignment; and the SE(3)-refined model's near-zero residual ($0.003°$ RMS) confirms convergence. At no point does the user need to guess—the residual pattern at each stage guides the next hypothesis.

In this case study, the Z_0^0-dominated residual correctly guided us to the SE(3) arm alignment, while alternative hypotheses (pre-warp, variable origin) were correctly rejected by the same diagnostic.

5.2 Limitations

1. **Single dataset.** Results are demonstrated on 10 stereo pairs from one Pycaso CMO microscope. Generalisation to other CMO designs, manufacturers, or magnification settings has not been

demonstrated and requires additional validation.

2. **Calibration residuals use the board points.** All ray-space and pixel-equivalent residuals are computed on the same ChArUco board points used for calibration. The independent surface check is profilometry of the specimen (§4.9); a separate *certified* 3D artefact at a known distance remains unavailable for the open Pycaso dataset.

3. **Depth scale requires acquisition metrology.** The specimen sanity check (§4.8) reveals a $\sim 30\,\%$ depth-only scale difference between the free-pose Zernike and CMO 26p reconstructions ($s_z \approx 0.70$). External profilometry shows that the free-pose Zernike solution over-amplifies the relief by $1.63\times$, whereas the same Zernike family reaches $1.04\times$ when the nominal translation-stage ladder is supplied as a near-fixed pose prior; the CMO 26p and Soloff references reach $1.10\times$ and $1.09\times$, respectively. This resolves the observed discrepancy as a pose-metrology ambiguity rather than a limitation of the Zernike basis. The result still depends on trusting the stage coordinates, and validation is limited to one specimen with point-wise lateral registration $\mathrm{cc} = 0.96$; it therefore does not certify an absolute depth uncertainty.

4. **Effective rayfield descriptors are gauge-sensitive.** The geometric descriptors (b, WD, f_{obj}, θ) are read from the rayfield under the transverse gauge $O \cdot d = 0$ and a fixed pinhole reference $f_x = 25600$. The working distance is linearly proportional to the initial f_x ($\pm 10\%$ fx $\rightarrow \pm 10\%$ WD, Section 4.7), and is *not* independently identifiable without an external scale reference.

5. **SE(3) translation parameters are not uniquely identifiable.** The rotation angles ($\sim 2.5°$, $\sim 3.7°$) are stable across optimisation runs, but the translation components trade off with the telecentric base parameters (WD, f_{obj}, b, principal point). Only the SE(3) rotation is a robust diagnostic; the translation is effective rather than absolute.

6. **The Zernike rayfield is a training residual.** Its $0.0007\,\mathrm{mm}$ ray-space RMS is a self-evaluation on the support points it was fitted to; the $0.47\,\mathrm{px}$ pixel RMS is a noise-floor estimate, not an independent measurement. The pixel RMS column in all tables provides the apples-to-apples comparison.

7. **Fixed pinhole reference.** The Zernike BA uses a fixed $f_x = 25600$ and principal point at image centre. Errors in this reference propagate to the rayfield estimates and the derived physical descriptors.

8. **Monochromatic effective model.** The rayfield and all derived physical models are identified from broad-spectrum white-light images and represent an effective model at the sensor's integrated spectral band (visible, $\sim 400\,\mathrm{nm}$ to $700\,\mathrm{nm}$). CMO microscopes carry residual chromatic aberration—both lateral (magnification variation with wavelength) and longitudinal (focus shift)—that is absorbed into the effective ray directions and origins per pixel. A wavelength-resolved rayfield $\mathcal{R}(u, v, \lambda)$ is a natural extension but requires narrow-band or hyperspectral acquisitions not available in the present dataset; it would disentangle geometric misalignment from chromatic effects and is noted as a future direction.

9. **Absolute metrological validation would require:** (a) manufacturer specifications for the actual microscope used, (b) an independent measurement of WD or baseline, or (c) a certified 3D

calibration object at a known distance. Item (a) is now partly available—the instrument is identified as a Leica M205C with a Planapo 1.0× objective (documented $WD = 61.5\,\mathrm{mm}$ [13]), and the reconstructed baseline agrees with it to $\approx 1\,\%$ via the chief-ray relation (17) (§4.5)—but this is a physical *consistency* check that still uses the datasheet WD as the scale anchor, not an independent measurement; items (b) and (c) remain unavailable for the open Pycaso dataset. The present validation thus establishes internal geometric consistency, agreement with the documented optical configuration, model interpretability, and descriptor stability on a real CMO dataset, but not absolute 3D metrological accuracy. A future validation on a calibrated 3D artefact or stepped depth standard will be required to quantify absolute depth accuracy and uncertainty propagation.

5.3 Generalisability

The methodology—measure with a flexible non-parametric basis, read physical descriptors, diagnose residual structure, build compact physical model, validate by BIC—is transferable to any non-standard optical instrument (endoscopes, Scheimpflug cameras, multi-camera arrays). The SE(3) arm alignment is specific to CMO microscopes, but the residual analysis that identified it is general.

5.4 Why the Two-Stage Decomposition Works

The central insight of this work is not a specific model architecture but a problem decomposition strategy. Direct bundle adjustment jointly estimates optical parameters θ and board poses η from 2D corners; for non-central optics, the Schur complement of the reduced camera system [33] reveals coupling norms of $c = 0.81$–0.98 depending on the diagnostic configuration, meaning optics and poses are nearly inseparable—the problem is severely ill-conditioned. By inserting the Zernike rayfield as an intermediate variable, the estimation splits into two sub-problems: (1) fit a flexible rayfield to corners with a constrained pose model, and (2) identify physical optics in ray space, where the coupling norm is 0.0 by construction because board poses are no longer estimated (pose-free model selection with BIC).

This decomposition is the mechanism behind all three empirical findings: the TPS pre-processing stabilises step (1) by cleaning the 2D input; the rayfield decouples step (2) from pose estimation; and the rayfield-initialised parameters lie close to the constrained-pose corner optimum, while also providing the correct basin and observability prior for the free-pose Schur-stabilised refinement.

6 Conclusion

We have demonstrated the first real-data validation of the StereoComplex non-central calibration pipeline on a CMO stereo microscope. The key results are:

- A compact 26-parameter physical model (quasi-telecentric CMO skeleton + per-channel SE(3) arm alignment) achieves $1.06\,\mathrm{px}$ reprojection (P50=$0.87\,\mathrm{px}$, P95=$1.84\,\mathrm{px}$) where standard OpenCV stereo calibration fails (> 300 px).

- The Zernike rayfield serves as a diagnostic instrument: residual modal analysis identifies missing degrees of freedom, and the operational usability score with reprojection guard correctly selects the usable model.

- Effective rayfield descriptors—baseline, working distance, effective axial length, convergence angle—are read directly from the measured rays (within the chosen rayfield gauge and reference scale), without fitting a physical model.

- On the independent coin specimen, the free-pose CMO–Zernike relief discrepancy is a single depth-only scale factor ($s_z \approx 0.70$), not a non-rigid error, and is fixed by acquisition metrology rather than by the calibration likelihood. External profilometry (§4.9) shows the CMO 26p and Soloff reliefs match the profilometric ground truth to $\approx 10\,\%$ ($1.10\times$ and $1.09\times$); supplying the nominal translation-stage ladder as a near-fixed hierarchical pose prior (§4.8.1) brings the Zernike rayfield to the same footing ($1.04\times$), without degrading the ray-space fit. The original $1.63\times$ free-pose over-amplification is therefore a missing pose-metrology prior, not an intrinsic limitation of the Zernike or non-central rayfield.

- The rayfield-initialised parameters are near-optimal for direct corner bundle adjustment ($17\,\%$ improvement), and the same rayfield provides a Schur-complement observability prior that stabilises the BA against optics–pose compensation.

- The accuracy–interpretability trade-off is quantified by comparison with the Soloff polynomial baseline (§4.6): a black-box polynomial achieves $0.18\,\mathrm{px}$ (80 params) but yields no physical descriptors; the Zernike rayfield reaches $0.41\,\mathrm{px}$ (93 params) under the 3D geometric consistency constraint ($0.23\,\mathrm{px}$ gap = cost of geometry); the compact CMO 26p model is identified at $1.06\,\mathrm{px}$ and, used as the initialisation for the free-pose Schur-stabilised bundle adjustment, refines to $0.28\,\mathrm{px}$ (within $0.1\,\mathrm{px}$ of the Soloff floor) with a physically interpretable parameterisation. We therefore do not merely trade accuracy for interpretability: the refined physical model is competitive in pixel accuracy with black-box regression while, unlike it, producing a physically meaningful calibration.

The CMO Pycaso case study validates a generalisable methodology—measure first, model second, diagnose by residual, refine by hypothesis testing—and establishes the Zernike rayfield as a multi-purpose instrument: it measures the rays, diagnoses the missing physical degree of freedom, initialises the bundle adjustment, and regularises it in two distinct senses—*structurally*, its 3D ray constraint resists overfitting and extrapolates better than an unconstrained polynomial (§4.6), and *statistically*, its Schur-complement spectrum supplies an observability prior that suppresses optics–pose compensation (§3.9). Notably, the axial-scale ambiguity left by the planar pose set (§4.8) never impeded this construction: the compact physical model was initialised, built, and—through the free-pose Schur-stabilised bundle adjustment—identified to $0.28\,\mathrm{px}$ regardless, precisely because the rayfield serves here as a model-construction instrument rather than a metrically self-sufficient calibration end-product. And where a metric depth scale is itself the goal, that same ambiguity is no longer an intrinsic barrier: §4.8.1 shows that transmitting the nominal translation-stage ladder to the rayfield bundle adjustment as a near-fixed hierarchical pose prior selects the metrically correct depth gain ($1.04\times$ the profilometry, against $1.63\times$ for the free-pose fit) without degrading the ray-space RMS—closing the depth-scale gauge problem from acquisition metrology alone, with the external profilometry serving only as an independent check rather than as the source of the scale. Code, data, and documentation are publicly available at `https://github.com/jeffwitz/StereoComplex`. A self-contained Zenodo archive (concept DOI 10.5281/zenodo.20444215) bundles the manuscript, figures, tables, numerical audit, reproducibility scripts, and all data files needed to rebuild the PDF and regenerate every figure. For exact reproducibility, the version submitted with this manuscript is archived as v8 (10.5281/zenodo.21025322); the concept DOI above is a stable landing page that always resolves to the latest version.

Prospective

Several directions extend this work. First, the methodology should be validated on additional CMO microscope designs and manufacturers to confirm generalisability beyond the single Pycaso instrument. Second, the Greenough stereo microscope family—with tilted optical axes rather than parallel sub-pupils—presents a harder identification problem where the telecentric model does not apply; the residual-guided construction method would need to discover a different geometric skeleton. Third, quantitative uncertainty propagation from ChArUco corner noise to effective-descriptor estimates (baseline, working distance) would strengthen the metrological claim. Fourth, deep learning-based corner detection could replace the hand-tuned Hessian+TPS pipeline, potentially improving the raw 2D input quality and further reducing the gauge instability. Finally, extending the Zernike rayfield to time-varying deformations (e.g., vibration, thermal drift) would enable *in-situ* calibration monitoring without dedicated targets.

Reproducibility Statement

The StereoComplex codebase (tag `v1.1-submission`, branch `develop`) and all artefacts required to reproduce these results are available at `https://github.com/jeffwitz/StereoComplex`. The raw Pycaso calibration images are available from `https://github.com/LaboratoireMecaniqueLille/Pycaso`. To reproduce the numerical results without access to the raw images, restart from the pre-computed intermediate state (`intermediate_state.npz`) which contains the already-detected, Hessian-completed, and TPS-denoised corner positions, the fitted Zernike rayfield, and the initial 26p model parameters. A self-contained Zenodo archive provides the manuscript, all figures, tables, numerical audit, reproducibility scripts, figure-generation manifests, and key data files (`intermediate_state.npz`, specimen correspondences, Schur diagnostic). For exact reproducibility, the version submitted with this manuscript is archived as v8 (10.5281/zenodo.21025322, a structured self-contained bundle—package source, reproduction scripts and key data, preserving the repository layout—plus the manuscript PDF); the concept DOI (10.5281/zenodo.20444215) is a stable landing page under which future versions are deposited.

A Numerical Reprojection for the CMO Model

The CMO telecentric model defines a forward mapping $\texttt{ray}(u, v) \to (O, d)$ from pixel coordinates to a 3D ray. Computing the 2D reprojection error requires the inverse: given a 3D point X^{cam} in the camera frame, find the pixel (u, v) whose ray passes through (or closest to) X^{cam}. For a non-central model, no closed-form inverse exists. We solve it numerically: for each observation, the predicted pixel is

$$(u_{\text{pred}}, v_{\text{pred}}) = \operatorname{argmin}_{(u,v)\in[0,W]\times[0,H]} \|X^{\text{cam}} - P(u, v; \theta)\|^2, \tag{19}$$

where $P(u, v; \theta)$ is the point on the ray $(O(u, v), d(u, v))$ closest to X^{cam}, computed as

$$P = O + \big((X^{\text{cam}} - O) \cdot d\big)\, d. \tag{20}$$

The objective is the squared point-to-ray perpendicular distance. The minimisation is performed with the L-BFGS-B algorithm [6], starting from the observed pixel $(u_{\text{obs}}, v_{\text{obs}})$, with bounds $[0, W-1] \times [0, H-1]$. Convergence tolerance is set to $\texttt{ftol}=10^{-12}$, $\texttt{gtol}=10^{-8}$. For the Pycaso dataset (10 frames, 165 corners, 2 channels; 3300 observations), the full reprojection computation takes approximately

35 seconds on a single CPU core.

The 2D reprojection residual for an observation is the vector $(u_{\text{pred}} - u_{\text{obs}},\ v_{\text{pred}} - v_{\text{obs}})$ in pixels. The 2D RMS reported in Section 3.9 and Table 2 is

$$\mathrm{RMS}_{\mathrm{2D}} = \sqrt{\frac{1}{N}\sum_{k=1}^{N}\left[(u_{\text{pred},k} - u_{\text{obs},k})^2 + (v_{\text{pred},k} - v_{\text{obs},k})^2\right]}, \tag{21}$$

where $N = n_{\text{frames}} \times n_{\text{corners}} \times 2$ is the total number of observations.

This numerical reprojection is used as a post-optimisation diagnostic: the bundle adjustment itself minimises the ray-space residual (:func:‘point_to_ray_residuals_cmo_se3‘ in the StereoComplex codebase), which is a forward evaluation and therefore faster and smoother. After convergence, the 2D reprojection error is computed from the optimised parameters to obtain the pixel-equivalent metric reported in the paper.

B Zernike Basis and Mode Indexing

The origin and direction fields of §3.1–3.2 are expanded on the real-valued Zernike basis as evaluated in the StereoComplex code. This appendix fixes the exact definition—convention, ordering, and the modes used—so that the modal language of the paper (the Z_0^0 piston diagnostic, second-order residuals) maps unambiguously onto basis functions and the rayfield fit is reproducible.

Definition. For integers $0 \le m \le n$ with $n - m$ even, the mode is

$$Z_n^m(\rho,\theta) = R_n^m(\rho)\,\Theta(\theta), \qquad \Theta(\theta) = \begin{cases} 1 & m = 0, \\ \cos m\theta & \text{cosine branch}, \\ \sin m\theta & \text{sine branch}, \end{cases} \tag{22}$$

with the standard Zernike radial polynomial

$$R_n^m(\rho) = \sum_{k=0}^{(n-m)/2} \frac{(-1)^k\,(n-k)!}{k!\,\left(\frac{n+m}{2} - k\right)!\,\left(\frac{n-m}{2} - k\right)!}\,\rho^{\,n-2k}. \tag{23}$$

Amplitude convention. The basis is used *unnormalised*: $R_n^m(1) = 1$, with *no* $\sqrt{n+1}$ / $\sqrt{2(n+1)}$ RMS (OSA/ANSI) factor. A fitted coefficient is therefore the mode's peak amplitude—in millimetres for the origin field, dimensionless for the direction perturbation. As a consequence the disk-mean square is $\langle (Z_n^m)^2 \rangle_{\text{disk}} = 1/(n+1)$ for $m = 0$ and $1/\big(2(n+1)\big)$ for $m \neq 0$ (not unity); the modes are an interpolation basis here, not an orthonormal frame.

Radial coordinate. Pixels map to polar coordinates by the diagonal (circumscribed-circle) normalisation of §3.2:

$$\rho = \frac{1}{\sqrt{2}}\left\|\left(\tfrac{2u}{W-1} - 1,\ \tfrac{2v}{H-1} - 1\right)\right\|, \qquad \theta = \operatorname{atan2}\!\left(\tfrac{2v}{H-1} - 1,\ \tfrac{2u}{W-1} - 1\right), \tag{24}$$

so the four image corners sit at $\rho = 1$ and every pixel satisfies $\rho \le 1$.

Ordering and modes. Modes are enumerated by increasing radial order n, then increasing azimuthal order m, the cosine branch before the sine branch; this is the column order of the design matrix. Table 14 lists them up to $n = 4$, covering the full BIC sweep of §3.1. An origin field of order o uses the first $N(o)$ modes and a direction field of order d the first $N(d)$ modes, with cumulative counts $N(0, 1, 2, 3, 4) = (1, 3, 6, 10, 15)$; each mode carries a 3-vector coefficient, so a channel has $3N(o) + 3N(d)$ rayfield parameters (e.g. o=0, d=2: $3+18 = 21$ per channel). The single subscript Z_m in §3.2 indexes this list; "Z_0^0" is the piston row, and a "second-order" residual refers to the $n = 2$ rows (defocus and astigmatism).

Table 14. Real Zernike modes in code order, up to radial order $n = 4$. "Branch" is the angular factor Θ; "j" is the 0-based column index of the design matrix. Classical names are given for reference only.

j	n	m	Branch	Classical name
0	0	0	—	piston
1	1	1	$\cos\theta$	tilt (x)
2	1	1	$\sin\theta$	tilt (y)
3	2	0	—	defocus
4	2	2	$\cos 2\theta$	astigmatism (0°)
5	2	2	$\sin 2\theta$	astigmatism (45°)
6	3	1	$\cos\theta$	coma (x)
7	3	1	$\sin\theta$	coma (y)
8	3	3	$\cos 3\theta$	trefoil
9	3	3	$\sin 3\theta$	trefoil
10	4	0	—	spherical
11	4	2	$\cos 2\theta$	secondary astigmatism
12	4	2	$\sin 2\theta$	secondary astigmatism
13	4	4	$\cos 4\theta$	quadrafoil
14	4	4	$\sin 4\theta$	quadrafoil

Source. The generator `zernike_modes` and the evaluator `eval_real_zernike` live in `src/stereocomplex/core/model_compact/zernike.py`; the diagonal pixel-to-disk map is in `src/stereocomplex/rayfields/zernike_origin_field.py` (`ZernikeOriginFieldConfig` with `normalization="diagonal_disk"`).

Author Contributions

J.-F. Witz conceived the study, developed the StereoComplex framework, implemented all methods, conducted the experiments, and wrote the manuscript.

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Code and Data Availability

- StereoComplex: `https://github.com/jeffwitz/StereoComplex` (Code: GPL-2.0-or-later. Documentation: CC BY-SA 4.0.)
- Pycaso dataset: `https://github.com/LaboratoireMecaniqueLille/Pycaso`
- Zenodo archive (manuscript + reproducibility assets): concept DOI 10.5281/zenodo.20444215; submitted version v8 10.5281/zenodo.21025322
- All generated artefacts: `docs/assets/pycaso_real_data/`
- Intermediate state (restart without raw images): `docs/assets/pycaso_real_data/intermediate_state.npz`

Declaration of competing interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author used AI-assisted tools (Claude, Anthropic; ChatGPT, OpenAI) in order to improve the language and readability of the manuscript. These tools were also used during development of the analysis code, as part of the research workflow documented in the Reproducibility Statement. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the scientific content, code, numerical results, figures, interpretations, and conclusions of the published article.

References

[1] H. Akaike. A new look at the statistical model identification. *IEEE Transactions on Automatic Control*, 19(6):716–723, 1974. doi:10.1109/tac.1974.1100705.

[2] A. Bartoli and P. Sturm. Structure-from-motion using lines: Representation, triangulation, and bundle adjustment. *Computer Vision and Image Understanding*, 100(3):416–441, 2005. doi:10.1016/j.cviu.2005.06.001.

[3] F. L. Bookstein. Principal warps: Thin-plate splines and the decomposition of deformations. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(6):567–585, 1989. doi:10.1109/34.24792.

[4] J.-Y. Bouguet. Camera calibration toolbox for MATLAB, 2008. URL: http://www.vision.caltech.edu/bouguetj/calib_doc/.

[5] D. C. Brown. Decentering distortion of lenses. *Photogrammetric Engineering*, 32(3):444–462, 1966.

[6] R. H. Byrd, P. Lu, J. Nocedal, and C. Zhu. A limited memory algorithm for bound constrained optimization. *SIAM Journal on Scientific Computing*, 16(5):1190–1208, 1995. doi:10.1137/0916069.

[7] F. Camposeco, T. Sattler, and M. Pollefeys. Non-parametric camera model calibration with splines. In *Proceedings of the German Conference on Pattern Recognition (GCPR)*, pages 285–296, 2016. doi:10.1007/978-3-319-45886-1_23.

[8] E. Caron, J.-F. Witz, C. Cuvier, A. Beaurain, V. Magnier, and A. El Bartali. PYCASO: Python module for calibration of cameras by Soloff's method. *SoftwareX*, 23:101440, 2023. doi:10.1016/j.softx.2023.101440.

[9] S. Garrido-Jurado, R. Muñoz-Salinas, F. J. Madrid-Cuevas, and M. J. Marín-Jiménez. Automatic generation and detection of highly reliable fiducial markers under occlusion. *Pattern Recognition*, 47(6):2280–2292, 2014. doi:10.1016/j.patcog.2014.01.005.

[10] M. D. Grossberg and S. K. Nayar. A general imaging model and a method for finding its parameters. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 108–115, 2001. doi:10.1109/iccv.2001.937611.

[11] S. Inoué and K. R. Spring. *Video Microscopy: The Fundamentals*. Plenum Press, New York, 2nd edition, 1997. doi:10.1007/978-1-4757-6925-8.

[12] J. Kannala and S. S. Brandt. A generic camera model and calibration method for conventional, wide-angle, and fish-eye lenses. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(8):1335–1340, 2006. doi:10.1109/tpami.2006.153.

[13] Leica Microsystems. *M125 C, M165 C, M205 C, M205 A Stereo Microscopes — Technical Data*. Leica Microsystems GmbH, 2019. Planapo 1.0× objective: working distance 61.5 mm.

[14] Z. Liu, J. Guo, W. Shi, X. Chen, and H. Xie. Generic distortion model for metrology under optical microscopes. *Optics and Lasers in Engineering*, 103:119–126, 2018. doi:10.1016/j.optlaseng.2017.12.006.

[15] C. Mei and P. Rives. Single view point omnidirectional camera calibration from planar grids. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, pages 3945–3950, 2007. doi:10.1109/robot.2007.364084.

[16] P. Miraldo and H. Araujo. Calibration of smooth camera models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 35(9):2091–2103, 2013. doi:10.1109/TPAMI.2012.258.

[17] R. J. Noll. Zernike polynomials and atmospheric turbulence. *Journal of the Optical Society of America*, 66(3):207–211, 1976. doi:10.1364/josa.66.000207.

[18] OpenCV. Open source computer vision library. URL: https://opencv.org.

[19] J. Plücker. On a new geometry of space. *Philosophical Transactions of the Royal Society of London*, 155:725–791, 1865.

[20] S. Ramalingam and P. Sturm. A unifying model for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 39(7):1309–1319, 2017. doi:10.1109/tpami.2016.2592904.

[21] S. Ramalingam, P. Sturm, and S. K. Nayar. A generic structure-from-motion framework. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 218–231, 2010. doi:10.1007/978-3-642-15552-9_16.

[22] F. J. Romero-Ramirez, R. Muñoz-Salinas, and R. Medina-Carnicer. Speeded up detection of squared fiducial markers. *Image and Vision Computing*, 76:38–47, 2018. doi:10.1016/j.imavis.2018.05.004.

[23] J. Sanderson. *Understanding Light Microscopy*. Wiley, 2019. doi:10.1002/9781118696736.

[24] D. Scaramuzza, A. Martinelli, and R. Siegwart. A toolbox for easily calibrating omnidirectional cameras. In *Proceedings of the IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5695–5701, 2006. doi:10.1109/iros.2006.282372.

[25] T. Schöps, V. Larsson, M. Pollefeys, and T. Sattler. Why having 10,000 parameters in your camera model is better than twelve. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 2535–2544, 2020. doi:10.1109/cvpr42600.2020.00261.

[26] H. W. Schreier, D. Garcia, and M. A. Sutton. Advances in light microscope stereo vision. *Experimental Mechanics*, 44(3):278–288, 2004. doi:10.1007/BF02427894.

[27] G. Schwarz. Estimating the dimension of a model. *Annals of Statistics*, 6(2):461–464, 1978. doi:10.1214/aos/1176344136.

[28] S. M. Soloff, R. J. Adrian, and Z. C. Liu. Distortion compensation for generalized stereoscopic particle image velocimetry. *Measurement Science and Technology*, 8(12):1441–1454, 1997. doi:10.1088/0957-0233/8/12/008.

[29] C. Strecha et al. On benchmarking camera calibration and multi-view stereo for high resolution imagery. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 1–8, 2008. doi:10.1109/cvpr.2008.4587706.

[30] P. Sturm and S. Ramalingam. A generic concept for camera calibration. In *Proceedings of the European Conference on Computer Vision (ECCV)*, pages 1–13, 2004. doi:10.1007/978-3-540-24671-8_1.

[31] B. Triggs, P. McLauchlan, R. Hartley, and A. Fitzgibbon. Bundle adjustment — a modern synthesis. In *International Workshop on Vision Algorithms*, pages 298–372, 1999. doi:10.1007/3-540-44480-7_21.

[32] S. Weber, N. Demmel, and D. Cremers. Power bundle adjustment for large-scale 3D reconstruction. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 281–289, 2023. doi:10.1109/CVPR52729.2023.00035.

[33] J.-F. Witz. StereoComplex: Non-central stereo calibration framework, 2025. URL: https://github.com/jeffwitz/StereoComplex.

[34] F. Zernike. Diffraction theory of the knife-edge test and its improved form, the phase-contrast method. *Monthly Notices of the Royal Astronomical Society*, 94:377–384, 1934. doi:10.1093/mnras/94.5.377.

[35] D. Zhang, M. Luo, and D. D. Arola. Displacement/strain measurements using an optical microscope and digital image correlation. *Optical Engineering*, 45(3):033605, 2006. doi:10.1117/1.2182108.

[36] Z. Zhang. A flexible new technique for camera calibration. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 22(11):1330–1334, 2000. doi:10.1109/34.888718.